



A reinforcement learning approach to network-level maintenance planning of co-located road and water infrastructures with interdependencies[☆]

Bingjie Wang^a, Hung Nguyen^b, Qing Lu^b, Qiong Zhang^b, Mingyang Li^{a,*}

^a Department of Industrial and Management Systems Engineering, University of South Florida, 4202 E. Fowler Avenue, Tampa, 33620, FL, United States

^b Department of Civil and Environmental Engineering, University of South Florida, 4202 E. Fowler Avenue, Tampa, 33620, FL, United States

ARTICLE INFO

Keywords:

Joint maintenance planning
Infrastructure interdependency
Network-wide decision-making
Reinforcement learning
Co-located deteriorating systems

ABSTRACT

Due to co-location and spatial proximity, deteriorating transportation infrastructure (TI) and water infrastructure (WI) are geographically interdependent, exhibiting complex interactions both within and across systems. Failures from degradation or maintenance disruptions in one system can impair its own performance and cascade into the other. However, many existing studies address long-term maintenance only at the component, subsystem, or single-system level, without fully accounting for interdependencies both within and across co-located infrastructures. This paper proposes a network-wide joint long-term maintenance planning framework that integrates traffic and hydraulic simulations to capture interdependencies within and across multiple infrastructure systems. A reinforcement learning-based optimization model is formulated to minimize the expected cumulative costs of co-located TI and WI systems over a long-term planning horizon while accounting for future uncertainties in the deterioration of roads and pipes. The model jointly quantifies both short-term and long-term economic impacts of joint maintenance decisions at network level by comprehensively investigating diverse topological relationships and interactions between co-located roads and pipes. The framework is further validated through a realistic case study in Ybor City, Tampa, and benchmarked against alternative maintenance strategies, demonstrating cost savings achieved through proactive, coordinated decision-making across interdependent TI and WI systems.

1. Introduction

Infrastructure systems are vital to economic activity and daily life, but both TI and WI face increasing challenges from aging and deterioration. US DOT-FHWA [1] reported that nearly 69.5% of the 440,504 miles of U.S. highways are in fair, poor, or very poor condition. Fan et al. [2] estimated that over 700 water pipes fail daily in North America, causing the loss of more than 2 trillion gallons of water annually and requiring about \$1 trillion in repairs. These failures drive up costs and disruptions, underscoring the need for strategic maintenance. In practice, many roads and water pipes are spatially co-located, presenting both risks and potential opportunities for infrastructure management. In terms of risks, disruptions in one system can cascade through its own network and propagate to the other. For instance, maintenance work on a road segment can restrict lanes and slow traffic, creating delays that spread through nearby corridors, while shutting down a water pipe can disrupt supply to users and strain surrounding pipes. Furthermore, pipe failures often require excavation, disrupting co-located roads and incurring significant traffic and

repaving costs [3]. In turn, heavy vehicle loads can accelerate deterioration of the buried pipes beneath, contributing to spatial variability in failure rates [4,5]. In terms of opportunities, co-location can enable cost savings through coordinated maintenance. Pericault et al. [6] showed that joint road–pipe maintenance can avoid redundant efforts, such as repeated repaving and traffic control in the sharing area, and facilitate resource sharing in labor, equipment, and materials, thereby shortening total maintenance time and reducing overall costs. These intertwined risks and opportunities highlight the need for a joint, cost-efficient maintenance planning framework for co-located roads and water pipes that holistically accounts for their complex interdependencies.

1.1. Literature review

Many infrastructure maintenance and intervention planning studies focus on a single critical infrastructure, such as road pavements [7], water distribution networks [8], or bridges [9,10], either at the component or system level. While these studies offer useful insights for

[☆] This article is part of a Special issue entitled: 'OR&AI maintenance' published in Reliability Engineering and System Safety.

* Corresponding author.

E-mail address: mingyangli@usf.edu (M. Li).

Nomenclature

Symbol	Definition
$i \in \mathcal{I}$	Index and set of road segments in the TI network
$j \in \mathcal{J}$	Index and set of pipe segments in the WI network
$t \in \mathcal{T}$	Time index; set of planning years
H	Length of the planning horizon in years
$s_{i,t}^r$	Condition state of road i at time t
S^r	Discretized set of road condition states
Ω^r	Worst condition state (failure) for roads
$s_{j,t}^w$	Aging state of pipe j at time t
S^w	Set of discrete pipe aging states
Ω^w	Failure state for pipes
$a_{i,t}^r$	Maintenance action for road i at time t
$a_{j,t}^w$	Maintenance action for pipe j at time t
$p_i^{r,0}(s' s)$	Transition probability of road i from s to s' (no maintenance)
$p_i^r(s' s, a)$	Road transition probability under maintenance action a
$q_i^r(s^{\text{int}} s, a)$	Intermediate state right after road maintenance
$p_j^{w,0}(s' s)$	Pipe aging transition from s to s' (no maintenance)
$p_j^w(s' s, a)$	Pipe transition under action a
$q_j^w(s^{\text{int}} s, a)$	Intermediate pipe state right after maintenance
$\lambda_{j,t}$	Pipe failure rate at time t
λ_{j,t_0}	Initial pipe failure rate at installation
$A(s)$	Breakage rate growth coefficient based on pipe age
$\mathbf{1}_{s \leq s'}$	Indicator function (1 if $s \leq s'$, 0 otherwise)
$n_{ss'}$	Number of transitions observed from s to s'
$\mathcal{P}^r, \mathcal{P}^w$	Sets of roads/pipes scheduled for maintenance
$\bar{\mathcal{P}}^r, \bar{\mathcal{P}}^w$	Sets of roads/pipes not scheduled for maintenance
$\mathcal{P}^{r,k}, \mathcal{P}^{w,k}$	Subsets based on co-location types ($k \in \{1, 2, 3\}$)
B_i^r, B_j^w	Indices of pipes/roads co-located with road i /pipe j
L_i^r, L_j^w	Length of road i or pipe j
$c_{i,t}^{\text{m},r}, c_{j,t}^{\text{m},w}$	Unit maintenance cost for road/pipe at t
α_i, α_j	Influence factor of road i or pipe j
$F^r(\cdot), F^w(\cdot)$	The intra-network disruptions estimated via simulations for TI and WI
$\mathbf{z}_i^r, \mathbf{z}_j^w$	Feature vectors used in simulators
TF_t	Average annual daily traffic (AADT) at year t
$c^{\text{ust},r}, c^{\text{ust},w}$	Unit cost of traffic delay/water shortfall
D_i^r, D_j^w	Duration of maintenance on road i or pipe j
$C_{i,t}^{\text{st},r-1}, C_{j,t}^{\text{st},w-1}$	Short-term cost for Category-I road/pipe
$\mathcal{U}_g^{\text{gm}}$	Set of components disabled during group g maintenance
$\tau_g^{\text{gm}}, \Delta t_g^{\text{gm}}$	End time and duration of group g maintenance
$c_{i,j,t}^{\text{rp}}$	Daily repaving cost for co-located road i and pipe j
$c_{i,j,t}^{\text{tc}}$	Daily traffic control cost for co-located components
$\Delta T_{i,j,t}^{\text{m}}$	Duration reduction from joint maintenance
$L_{i,j}^{\text{co}}$	Shared length of co-located road i and pipe j
$C_{i,t}^{\text{lt},r}$	Long-term cost of road i from year t to $t+1$
T^e	Evaluation time period for long-term cost
$c^{\text{ult},r}$	State-dependent annualized unit cost of long-term road user
$C_{j,t}^{\text{lt},w}$	Long-term cost of pipe j from year t to $t+1$
c^{leak}	Unit leakage cost for pipe j
$c_{j,k}^{\text{break}}$	Cost of break for pipe j in subsegment type k
$L_{j,k}^w$	Length of type- k subsegments of pipe j
$\lambda_{j,k,t}$	Failure rate of pipe j type- k segment at t
$\mathcal{K}$	Set of pipe subsegment types (I, II, III)
$\mathbf{s}_t^r, \mathbf{s}_t^w$	State vectors of roads and pipes at time t
$\mathbf{a}_t^r, \mathbf{a}_t^w$	Action vectors of roads and pipes at time t
$S_{\text{joint}}^r, S_{\text{joint}}^w$	Joint state spaces of roads and pipes
$\pi(\cdot s)$	Maintenance policy mapping state to action
$J_t^\pi(\mathbf{s}, \mathbf{a})$	Expected cumulative cost from year t under policy π
$C_t(\mathbf{s}, \mathbf{a})$	Stage cost at time t
γ	Discount factor for future costs

$p(s' s, a)$	State transition probability function
δ	Temporal difference (TD) error in DQN
$\mathcal{B}$	Experience replay buffer
θ, θ^-	Parameters of DQN current and target networks

infrastructure-specific planning, they address isolated networks and do not consider dependencies within or across infrastructure systems. Dekker et al. [11] classified multi-component maintenance dependencies into three types—economic, structural, and stochastic. Economic dependence arises when joint maintenance leads to cost savings through economies of scale, or conversely, when simultaneous downtime is undesirable and requires staggered scheduling. Structural dependence refers to physical integration among components, where maintaining one component necessitates maintenance of others. Stochastic dependence occurs when the condition or failure of one component affects the failure probabilities of others, often due to shared deterioration processes or common-cause failures. Building on this typology, several studies have incorporated such dependencies into single-system maintenance models, capturing structural [12–14], stochastic [15–18], or economic [12,14,18,19] interactions for maintenance modeling and optimization.

Beyond maintenance literature for a single infrastructure system with intra-system dependencies among components, several recent works have examined the challenges of coordinating intervention planning across multiple interdependent infrastructure systems. These works recognize that maintenance actions in one system can significantly impact the performance, cost, or serviceability of other interconnected systems. To conceptualize such inter-system dependencies, Rinaldi et al. [20] proposed a widely adopted framework that classifies them into four types, namely physical, cyber, geographic and logical interdependencies. This typology offers a broad, system-level perspective, emphasizing bidirectional linkages and cascading effects across interconnected systems. Building on that, Kammouh et al. [21] studied interdependent railway, highway, and water networks, focusing on geographic linkages arising from spatial proximity. They proposed a deterministic MILP-based maintenance framework to coordinate intervention groupings and minimize total costs. But they did not consider deterioration uncertainties of infrastructure. Hu and Zhang [22] proposed a bi-level optimization framework that minimizes traffic disruption by coordinating WI repairs with transportation network operations, considering geographic interdependence between the two systems. But this framework does not incorporate dynamic degradation or address long-term costs. Nguyen et al. [23] studied co-located road sections and water pipes, focusing on geographic linkages. They proposed a joint maintenance planning framework that incorporates deterioration uncertainties, and demonstrates the cost-saving benefits of joint maintenance over separate planning. However, it focused on the component-level maintenance planning without considering intra- or inter-system linkages of two networks. Beyond long-term intervention planning, a growing body of research has focused on short-term restoration strategies for interdependent infrastructure systems, particularly under natural hazard scenarios. For instance, Ilalokhoin et al. [24] studied interdependent rail and power systems, focusing on physical dependencies where traction power failures disrupt rail operations under different failure and recovery scenarios. Kong et al. [25] studied physically interdependent electric, gas, and oil transmission networks. They developed a resilience optimization model to enhance system performance under hazard scenarios. Xu et al. [26] studied interdependent water and power infrastructure systems with bidirectional physical dependencies in a post-disaster recovery context. They developed a restoration framework using deterministic and two-stage stochastic models to optimize repair sequencing. Pei et al. [27] proposed an ANN-based recovery optimization framework for interdependent transportation and healthcare systems affected by earthquakes, focusing on geographic and functional dependencies that

influence medical accessibility during recovery. These studies have significantly advanced the understanding of interdependencies across infrastructure systems. However, they primarily address short-term restoration under hazards, rather than long-term maintenance planning under normal operating conditions.

The long-term joint maintenance planning of co-located TI and WI involves several layers of complexity. First, both intra- and inter-system dependencies must be carefully considered. Intra-system dependencies arise when maintenance or failure of one component, such as a road segment or pipe, affects neighboring components and propagates through the network. For example, blocking a road for maintenance may reroute traffic, causing congestion and cascading delays, while isolating a water pipe for maintenance can disrupt supply to downstream users and increase stress on adjacent pipes. Inter-system dependencies emerge from spatial co-location: pipe failures require excavation that disrupts roads and incurs significant traffic and repaving costs [28, 29]. In turn, heavy vehicle loads can accelerate deterioration of the buried pipes beneath, contributing to higher failure rates [4,5]. Second, economic dependencies compound these effects, as joint planning can reduce costs through shared operations such as repaving and traffic control. Third, the topological arrangements of roads and pipes vary widely across networks [30], from independent assets to closely coupled ones requiring simultaneous intervention. This variation complicates coordinated planning, as both short-term impacts (e.g., traffic delays and service disruptions) and long-term benefits (e.g., user cost reduction, extended performance) must be weighed. Moreover, effective long-term planning must account for uncertainties from stochastic road deterioration and sudden pipe failures under varied maintenance decisions.

To enable joint maintenance planning for co-located TI and WI with large number of interconnected components subject to uncertain road deterioration and pipe failures, reinforcement learning (RL) offers a promising framework. Recent studies have increasingly applied RL to infrastructure maintenance planning, primarily within a single infrastructure system. For instance, Zhang and Si [31] presented a RL model for maintaining a single system with multiple components facing competing failure risks, aiming to minimize long-term costs under economic and stochastic dependencies. Marugán et al. [32] developed a RL-based agent to optimize maintenance timing for a system with continuous degradation. Mohammadi and He [33] developed a RL framework for maintaining a rail infrastructure system under uncertainty and budget constraints, optimizing renewal and maintenance scheduling. Hamida and Goulet [34] proposed a hierarchical RL model for maintaining a bridge network, designed to manage large state-action spaces arising in long-term maintenance planning. Saifullah et al. [35] applied a multi-agent RL approach to a transportation system composed of bridges and pavements. Andriotis and Papakonstantinou [36] developed a RL method for maintaining a multi-component system under partial observability and resource constraints. Only a few studies explored RL applications at the network level by investigating dependencies among components. For instance, Morato et al. [37] extended RL-based maintenance to a single deteriorating system by modeling stochastic intra-system dependencies using dynamic Bayesian networks. Van Remmerden et al. [38] introduced a multi-objective RL approach for optimizing maintenance policies in a single infrastructure system, considering the stochastic dependencies in the system. While RL shows strong potential for optimizing complex maintenance planning, most studies still treat systems in isolation or only consider intra-system dependencies. Few works have addressed joint maintenance across multiple co-located systems by simultaneously accounting for both intra- and inter-system dependencies.

1.2. Contribution of the paper

To address the aforementioned gaps in the existing literature, we propose a long-term, network-wide joint maintenance planning model under an RL framework. The proposed model explicitly incorporates both intra- and inter-system dependencies between TI and WI systems, accounting for uncertainties in road deterioration and pipe failures. It also considers diverse topological relationships among TI and WI components, enabling more realistic, coordinated, and cost-effective maintenance decision-making. This includes capturing the mutual influence of maintenance actions across the two systems and quantifying their effects on traffic disruption and pipe failure rates. Second, we employ traffic and hydraulic microsimulations to evaluate how maintenance decisions propagate within road and pipe networks, thereby quantifying intra-system dependencies among components. Third, we integrate these microsimulations with the mathematical maintenance model under the RL framework to comprehensively estimate long-term cost components, accounting for stochastic road deterioration and random pipe failures. We also explicitly capture the economic dependencies of joint maintenance by quantifying both short- and long-term user costs from maintenance activities. Finally, we derive near-optimal, cost-effective maintenance policies under the DRL-based framework and demonstrate the approach through a realistic case study in Ybor City, Tampa. Results show that our joint maintenance strategy outperforms benchmark strategies that neglect interdependencies between the two systems. The main contributions of this paper are as follows:

1. We propose a long-term maintenance modeling approach under a reinforcement learning framework for two interdependent systems of co-located TI and WI, capturing both intra- and inter-system interdependencies as well as the uncertainties of road deterioration and pipe failures.
2. We model the inter-system geographical interdependencies between co-located TI and WI, as well as the economic interdependencies of their maintenance decisions under diverse topological relationships.
3. We comprehensively quantify the short-term and long-term impacts of joint maintenance decisions within and across infrastructure systems by integrating mathematical modeling with traffic and hydraulic microsimulations.
4. We develop a realistic testbed in Ybor City, Tampa, to showcase the cost-saving benefits of the proposed joint maintenance approach.

The structure of this paper is as follows: Section 2 presents the methodology, including an overview of the framework (Section 2.1), the modeling of road degradation performance for TI maintenance (Section 2.2), and pipe failure modeling for WI maintenance (Section 2.3). Sections 2.4 and 2.5 formulate the short-term and long-term cost models, respectively, accounting for the complex dependencies both within and between TI and WI networks. Section 2.6 introduces the optimization framework for joint maintenance planning using a DRL approach. Section 3 provides a case study to demonstrate the applicability of the proposed framework, and Section 4 concludes the paper by summarizing the key findings and contributions.

2. Methodology

2.1. Overview

This paper proposes a network-level, long-term joint maintenance framework for co-located TI and WI by simultaneously considering both their intra- and inter-system dependencies among components (e.g., roads and pipes). To clearly describe the proposed methodology, we adopt a bottom-up model formulation, as summarized in Fig. 1.

At the component level, we model the stochastic deterioration of individual roads and pipes, accounting for spatial variability and different maintenance actions (Sections 2.2–2.3). At the dependency level, Section 2.4 models the short-term maintenance costs of co-located roads and pipes, accounting for their diverse topological relationships. Section 2.5 then models the long-term effects of joint maintenance actions together with the resulting deterioration conditions of roads and pipes. In both sections, traffic and hydraulic simulations are integrated to quantify intra- and inter-system dependencies within and across networks. At the system level, Section 2.6 formulates a network-wide, long-term MDP-based maintenance planning model that aggregates component states and actions into a system-wide optimization problem, and presents the solution approach using DRL. The objective is to determine joint maintenance policies that minimize expected cumulative costs over the entire planning horizon under uncertainty. Due to the high dimensionality of both the state and action spaces, DRL is employed to find optimal decisions. To validate the framework, Section 3 presents a realistic case study in Ybor City, Tampa.

2.2. Degradation performance modeling of roads (TI)

To effectively model the stochastic deterioration of infrastructure components, we adopt a discrete-time Markov decision process (MDP). In this study, decisions are made at the beginning of each time epoch, and we assume the Markov property holds, meaning that the condition of a component depends only on its state in the previous epoch and the maintenance action applied (e.g., repair, do nothing). Such assumptions are widely used in infrastructure maintenance modeling [9,10,31,32]. Furthermore, because our focus is on long-term planning under normal operating conditions — relatively stable periods with similar environmental and usage patterns — we assume stationary transition dynamics for each component.

Road performance deteriorates stochastically over time due to factors such as traffic loads, material properties, design, and climate. To capture these deterioration uncertainties, we represent the condition of each road segment by a discrete state variable. Considering a TI network composed of multiple road segments, we define s_i as the degradation state of road i , where $i \in \mathcal{I} = \{1, 2, \dots, I\}$. The explicit index i is used because deterioration impacts overall network performance differently depending on the segment's location. Road conditions are commonly assessed using the International Roughness Index (IRI), which quantifies surface roughness as cumulative vertical displacement per unit distance [39]. Higher IRI values indicate more deteriorated surfaces. In this study, IRI values are discretized into finite states to support condition-based maintenance planning. Over an H -year planning horizon, annual maintenance decisions are made at the beginning of each year, $t \in \mathcal{T} = \{1, 2, \dots, H\}$. The state space of each road segment is defined as $\mathcal{S}^r = \{0, 1, \dots, \Omega^r\}$, where 0 represents perfect condition and Ω^r denotes the failure state requiring complete resurfacing or reconstruction [23]. If a road reaches the failure state, a perfect maintenance (PM) action resets its condition to state 0. In the absence of maintenance, deterioration follows the transition probability matrix $P^{r,0} = [p_i^{r,0}(s'|s)]$, where $p_i^{r,0}(s'|s)$ is the probability that segment i moves from state s to s' between two consecutive years. This deterioration process is formally expressed in Eq. (1):

$$p_i^{r,0}(s'|s) = \Pr(s_{i,t+1}^r = s' | s_{i,t}^r = s, a_{i,t}^r = \text{DN}) \times \mathbf{1}_{s \leq s'},$$

$$\text{for } i \in \mathcal{I}, t \in \mathcal{T} \setminus \{H\}; s, s' \in \mathcal{S}^r, \quad (1)$$

where $\mathbf{1}_{s \leq s'}$ is an indicator function ensuring that transitions only occur to equal or worse states due to road deterioration in the absence of maintenance. Based on the observed IRI assessment data collected in the field, maximum likelihood estimation (MLE) is employed to estimate the unknown transition probability quantities in $P^{r,0} = [p_i^{r,0}(s'|s)]$. The MLE estimate is given in Eq. (2):

$$\hat{p}_i^{r,0}(s'|s) = \frac{n_{ss'}}{n_s} = \frac{n_{ss'}}{\sum_{s'=0}^{\Omega^r} n_{ss'}}, \quad \text{for } i \in \mathcal{I}; \quad (2)$$

where

$$q_j^w(s^{\text{int}}|s, a_{j,t}^w) = \begin{cases} \frac{1}{s-1} & \text{if } a_{j,t}^w = \text{MM and } s^{\text{int}} \in S^w \setminus \{0, s\}, \\ 1 & \text{if } a_{j,t}^w = \text{PM and } s^{\text{int}} = 0, \\ 0 & \text{otherwise.} \end{cases}$$

As shown in Eq. (6), when no maintenance is taken ($a_{j,t}^w = \text{DN}$), pipes progress naturally according to $p_j^{w,0}(s'|s)$. After MM or PM, pipes are restored to an intermediate state s^{int} before degrading naturally until the next maintenance. Similar to roads, pipe maintenance affects not only the repaired pipe but also neighboring pipes, for example by causing temporary water supply shortfalls. In addition, excavation for pipe repairs can disrupt co-located roads, introducing interdependencies with TI. These intra- and inter-system dependencies are accounted for explicitly in later sections.

2.4. Short-term costs modeling for co-located roads and pipes

We distinguish between short-term costs, which capture the immediate impacts of maintenance actions, and long-term costs, which capture the cumulative consequences over the planning horizon. This subsection models short-term costs, including both direct maintenance and immediate user costs. At each decision epoch t , direct maintenance costs consist of repair expenses and traffic control costs. Repair expenses include construction setup, materials, and labor for operations such as milling and patching for roads or excavation and lining for pipes. Traffic control costs cover the labor, equipment, and signage required to manage disruptions during these activities. Indirect user costs arise from traffic delays due to lane closures and water service interruptions from outages. These temporary effects diminish once repairs are completed and are therefore categorized as short-term cost components. Long-term costs, such as those from poor infrastructure conditions or deferred maintenance, are addressed in the next subsection.

Because of the topological relationships between roads and pipes, indirect short-term user costs vary by scenario. If roads and pipes are not co-located, repairs affect only their respective users. When co-located, pipe repairs often require road excavation, which disrupts both traffic and water service even without concurrent road work. Overlapping repairs on multiple co-located components further intensify these disruptions due to geographical dependencies. Conversely, when both roads and pipes are repaired jointly, there are opportunities to reduce costs. Joint repairs require only a single traffic control setup, avoiding redundant expenditures, and they eliminate repeated repaving on shared road areas, which shortens the overall maintenance duration and lowers traffic interruption costs.

To capture the short-term cost effects of road–pipe co-location, we classify roads and pipes based on their maintenance actions and co-location scenarios. Roads are first divided into two mutually exclusive subsets: $\mathcal{P}^r := \{i : a_{i,t}^r \neq \text{DN}\}$, the set of roads scheduled for repair, and $\bar{\mathcal{P}}^r := \{i : a_{i,t}^r = \text{DN}\}$, the set of roads with no repair action. For each road $i \in \mathcal{I}$, let $\mathcal{B}_i^r \subseteq \mathcal{J}$ denote the index set of co-located pipes. Similarly, for each pipe $j \in \mathcal{J}$, let $\mathcal{B}_j^w \subseteq \mathcal{I}$ denote the index set of co-located roads. The set $\mathcal{P}^r$ is further decomposed into three subsets: $\mathcal{P}^{r,1} := \{i : i \in \mathcal{P}^r, \mathcal{B}_i^r = \emptyset\}$, $\mathcal{P}^{r,2} := \{i : i \in \mathcal{P}^r, \mathcal{B}_i^r \neq \emptyset, \mathcal{B}_i^r \cap \mathcal{P}^w = \emptyset\}$, and $\mathcal{P}^{r,3} := \{i : i \in \mathcal{P}^r, \mathcal{B}_i^r \neq \emptyset, \mathcal{B}_i^r \cap \mathcal{P}^w \neq \emptyset\}$. These represent roads that (i) do not have co-located pipes, (ii) have co-located pipes but none under repair, and (iii) have co-located pipes with at least one also under repair. Pipes are classified in the same way with respect to their co-located roads. Table 1 and Fig. 2 summarize these definitions and provide illustrative examples. Based on these subsets, we define three categories of components according to the short-term user costs they incur. The cost models for each category are presented below.

The first group, referred to as “Category-I”, includes roads or pipes whose maintenance affects only their own system (TI or WI) without impacting the other. Roads in $\mathcal{P}^{r,1}$, which have no co-located pipes

(e.g., R_2 in Fig. 2), and roads in $\mathcal{P}^{r,2}$, which have co-located pipes but none under repair (e.g., R_1 with $\bar{P}_1$), are both classified as Category-I. For a Category-I road i , the short-term cost at time t is expressed as:

$$C_{i,t}^{\text{st},r-I} = c_{i,t}^{\text{m},r}(s_{i,t}^r, a_{i,t}^r) \cdot L_i^r \\ + \alpha_i F^r(s_{i,t}^r, a_{i,t}^r, (s_{i',t}^r, a_{i',t}^r)_{i' \in \mathcal{I} \setminus \{i\}}, \mathbf{z}_t^r) \cdot \text{TF}_i \\ \cdot c^{\text{ust},r} \cdot D_i^r(s_{i,t}^r, a_{i,t}^r), \quad \forall i \in \mathcal{P}^{r,1} \cup \mathcal{P}^{r,2}, \quad (7)$$

where the short-term cost for road i at time t , $C_{i,t}^{\text{st},r-I}$, consists of two main terms. The first term, $c_{i,t}^{\text{m},r}(s_{i,t}^r, a_{i,t}^r) \cdot L_i^r$, represents the direct maintenance cost. It equals the unit maintenance cost, which depends on the road's condition and chosen action, multiplied by the road length L_i^r . Poorer road conditions or more complex maintenance actions result in higher unit costs due to increased workload. In the second term, $\alpha_i F^r(\cdot)$, captures daily traffic delays caused by maintenance on road i . These delays depend not only on the state and action of road i but also on those of other roads, $(s_{i',t}^r, a_{i',t}^r)_{i' \in \mathcal{I} \setminus \{i\}}$. Maintenance on a single road can propagate delays through detours and congestion elsewhere in the network. To capture these interdependencies, we employ SUMO [44], a microscopic traffic simulation tool that models vehicle movements and routing dynamics under various traffic conditions. The function $F^r(\cdot)$ represents the network-wide traffic impact of maintaining road i together with other concurrently maintained roads. To isolate the contribution of road i , this aggregate impact is scaled by its influence factor α_i , which reflects the relative importance of road i within the network. In addition to the states and actions of all roads at time t , the simulation considers additional factors $\mathbf{z}_t^r$, including network configuration, vehicle characteristics, and routing decisions. These inputs allow $F^r(\cdot)$ to provide a realistic representation of traffic conditions. Finally, TF_i denotes the Average Annual Daily Traffic (AADT) in year t , $c^{\text{ust},r}$ is the unit user cost per vehicle-day of delay, and D_i^r is the maintenance duration for road i , determined by its state and chosen action.

Pipes in $\mathcal{P}^{w,1}$ (e.g., P_3) belong to Category-I because they have no co-located roads. For a Category-I pipe j , the short-term cost at time t is expressed as:

$$C_{j,t}^{\text{st},w-I} = c_{j,t}^{\text{m},w}(s_{j,t}^w, a_{j,t}^w) \cdot L_j^w \\ + \alpha_j F^w(s_{j,t}^w, a_{j,t}^w, (s_{j',t}^w, a_{j',t}^w)_{j' \in \mathcal{J} \setminus \{j\}}, \mathbf{z}_t^w) \\ \cdot c^{\text{ust},w} \cdot D_j^w(s_{j,t}^w, a_{j,t}^w), \quad \forall j \in \mathcal{P}^{w,1}. \quad (8)$$

where the total short-term cost for pipe j , $C_{j,t}^{\text{st},w-I}$, has two components. The first term, $c_{j,t}^{\text{m},w}(s_{j,t}^w, a_{j,t}^w) \cdot L_j^w$, represents the direct maintenance cost, equal to the condition- and action-dependent unit cost multiplied by the pipe length L_j^w . Similar to roads, pipes in worse condition or requiring more extensive repairs incur higher unit costs due to additional labor, material, and operational demands. The second term, $\alpha_j F^w(\cdot)$, accounts for daily service disruptions caused by maintenance on pipe j . Similar to roads, there are intra-system dependencies in WI: repairs on one pipe can affect neighboring pipes and reduce overall performance, leading to water shortfalls and unmet demand. These effects are quantified using a hydraulic simulation model developed in WaterGEMS [45], which captures the impact of suspending pipe j together with other concurrently repaired pipes. The model also incorporates additional factors $\mathbf{z}_t^w$, such as pipe characteristics, hydraulic constraints, and network topology [46]. Finally, $c^{\text{ust},w}$ is the unit cost of water service disruption per gallon, and D_j^w is the maintenance duration for pipe j , determined by its state and repair action.

The second category, referred to as “Category-II”, includes components where maintenance disrupts both TI and WI, even though only one side is under repair. In this case, a component has co-located infrastructure from the other system that is not undergoing maintenance. For example, pipes in subset $\mathcal{P}^{w,2}$ (e.g., P_4) fall into this category: they are scheduled for maintenance and have co-located roads, so the intervention disrupts both the water supply and traffic flow. However,

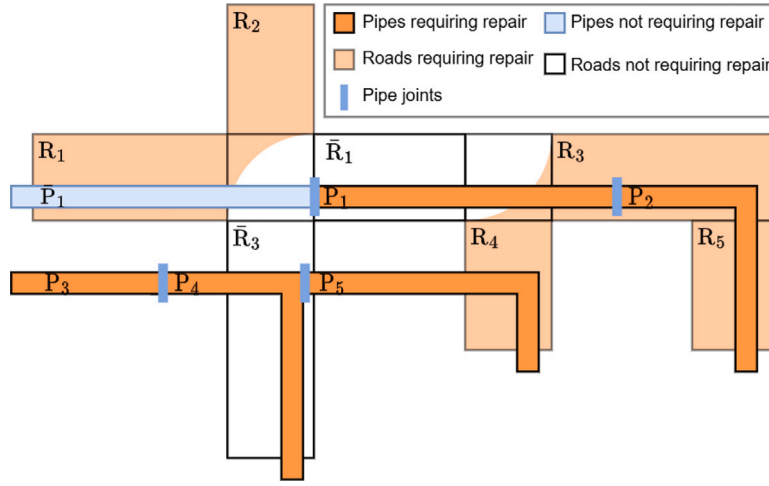


Fig. 2. Illustrative diagram of pipes and roads under different co-location and maintenance scenarios.

Table 1

Notations for roads and pipes with varied topological relationships.

Road			
Set	Subset	Definition and formulation	Example (Fig. 2)
Roads scheduled to repair: $\mathcal{P}^r := \{i : a_{i,t}^r \neq \text{DN}\}$ e.g. R_1, R_2, R_3, R_4, R_5	$\mathcal{P}^{r,1}$	Roads without co-located pipes: $\mathcal{P}^{r,1} := \{i : i \in \mathcal{P}^r, B_i^r = \emptyset\}$	R_2
	$\mathcal{P}^{r,2}$	Roads with co-located pipes, but all co-located pipes are not scheduled to be repaired: $\mathcal{P}^{r,2} := \{i \in \mathcal{P}^r, B_i^r \neq \emptyset, B_i^r \cap \mathcal{P}^w = \emptyset\}$	R_1
	$\mathcal{P}^{r,3}$	Roads with co-located pipes, and at least one also needs repair: $\mathcal{P}^{r,3} := \{i \in \mathcal{P}^r, B_i^r \neq \emptyset, B_i^r \cap \mathcal{P}^w \neq \emptyset\}$	R_3, R_4, R_5
Roads not scheduled to repair: $\bar{\mathcal{P}}^r := \{i : a_{i,t}^r = \text{DN}\}$			$\bar{R}_1, \bar{R}_2$
Pipe			
Set	Subset	Definition and formulation	Example (Fig. 2)
Pipes scheduled to repair: $\mathcal{P}^w := \{j : a_{j,t}^w \neq \text{DN}\}$ e.g. P_1, P_2, P_3, P_4, P_5	$\mathcal{P}^{w,1}$	Pipes without co-located roads: $\mathcal{P}^{w,1} := \{j : j \in \mathcal{P}^w, B_j^w = \emptyset\}$	P_3
	$\mathcal{P}^{w,2}$	Pipes with co-located roads, but all co-located roads are not scheduled to be repaired: $\mathcal{P}^{w,2} := \{j \in \mathcal{P}^w, B_j^w \neq \emptyset, B_j^w \cap \mathcal{P}^r = \emptyset\}$	P_4
	$\mathcal{P}^{w,3}$	Pipes with co-located roads, and at least one also needs repair: $\mathcal{P}^{w,3} := \{j \in \mathcal{P}^w, B_j^w \neq \emptyset, B_j^w \cap \mathcal{P}^r \neq \emptyset\}$	P_1, P_2, P_5
Pipes not scheduled to repair: $\bar{\mathcal{P}}^w := \{j : a_{j,t}^w = \text{DN}\}$			$\bar{P}_1$

since the co-located roads are not undergoing maintenance, no cost-sharing from joint maintenance is considered. For a Category-II pipe j , its short-term cost at year t due to maintenance action $a_{j,t}^w$ is given by:

$$\begin{aligned}
 C_{j,t}^{\text{st},w-\text{II}} &= c_{j,t}^{\text{m},w}(s_{j,t}^w, a_{j,t}^w) \cdot L_j^w \\
 &+ \alpha_j F^w(s_{j,t}^w, a_{j,t}^w, (s_{j',t}^w, a_{j',t}^w)_{j' \in J \setminus \{j\}}, \mathbf{z}_t^w) \cdot c^{\text{ust},w} \cdot D_j^w(s_{j,t}^w, a_{j,t}^w) \\
 &+ \sum_{i \in B_j^w} \alpha_i F^r(s_{i,t}^r, a_{i,t}^r, (s_{i',t}^r, a_{i',t}^r)_{i' \in I \setminus \{i\}}, \mathbf{z}_t^r) \cdot \text{TF}_i \\
 &\cdot c^{\text{ust},r} \cdot D_i^r(s_{i,t}^r, a_{i,t}^r), \quad \forall j \in \mathcal{P}^{w,2},
 \end{aligned} \quad (9)$$

where the total short-term cost for a Category-II pipe consists of three components. The first term represents the direct maintenance cost. The second captures the water user cost of the overall WI network due to the repair of pipe j , evaluated using the hydraulic simulation model described earlier for Category-I pipes. The third term, $\sum_{i \in B_j^w} \alpha_i F^r(\cdot) \cdot \text{TF}_i \cdot c^{\text{ust},r} \cdot D_i^r(s_{i,t}^r, a_{i,t}^r)$, accounts for the indirect road user costs caused by pipe construction. It represents the cumulative impact on all co-located roads in the set B_j^w , since pipe maintenance disrupts traffic flow even when no road repair is scheduled. The extent of this disruption

depends on TI-WI interactions, such as detours and congestion from road-space occupation during pipe repairs, and is quantified using traffic simulation.

The third category of roads or pipes, referred to as “Category-III”, includes cases where the road (or pipe) scheduled for maintenance has at least one co-located pipe (or road) that also requires repair. All roads in this category belong to the set $\mathcal{P}^{r,3}$, while pipes belong to $\mathcal{P}^{w,3}$. To reduce overall costs and improve efficiency, Category-III components are grouped for joint maintenance according to their spatial connectivity, which allows coordinated resource use and avoids redundant work. For example, in Fig. 2, $\mathcal{P}^{r,3}$ includes roads R_3, R_4, R_5 , while $\mathcal{P}^{w,3}$ includes pipes P_1, P_2, P_5 . These form two connected groups: $\{R_3, R_5, P_1, P_2\}$ and $\{R_4, P_5\}$, where components are modeled as vertices and co-location relationships as edges. By performing joint maintenance on such groups, redundant repaving and traffic control are eliminated, resources and workforce can be shared, and total construction duration is shortened compared to conducting repairs separately [23].

To capture these efficiency gains, our framework allows road and pipe maintenance teams to work simultaneously on interconnected TI and WI components within a group. Construction on all components

in a group is assumed to occur concurrently, enabling resource sharing and improved allocation. At the beginning of each maintenance phase at time t , Category-III components are grouped and then repaired sequentially, one group at a time. Within each group, multiple roads and pipes are maintained concurrently to capture joint efficiencies. Once all Category-III groups are completed, individual maintenance is carried out for Category-I and Category-II components. In this stage, due to capacity limits, at most one road and one pipe can be under repair at the same time. When a road or pipe finishes maintenance, the next scheduled component begins immediately, and this process continues until all scheduled components are repaired.

During the group maintenance stage, repairs are carried out sequentially, with groups processed in a random order. Let $\mathcal{U}_g^{\text{gm}}$ denote the set of all components disabled during the maintenance of the g th group within the interval $[\tau_{g-1}^{\text{gm}}, \tau_g^{\text{gm}}]$, where g ranges from 1 to the total number of groups. This set includes both components directly under repair and those indirectly affected, such as roads blocked due to pipe work. For example, in Fig. 2, all Category-III components can be divided into two groups. The first group, $\mathcal{U}_1^{\text{gm}}$, is blocked during $[\tau_0^{\text{gm}}, \tau_1^{\text{gm}}]$, and the second group, $\mathcal{U}_2^{\text{gm}}$, during $[\tau_1^{\text{gm}}, \tau_2^{\text{gm}}]$. These aggregated durations are shorter than repairing the grouped components individually in sequence. As an illustration, $\mathcal{U}_1^{\text{gm}}$ includes directly maintained components R_3, R_5, P_1 , and P_2 , as well as road $\bar{R}_1$, which is not scheduled for repair but is blocked due to its co-located pipe P_1 .

The total group maintenance cost for Category-III components at time t , based on group maintenance durations, is given by:

$$\begin{aligned} & \sum_{i \in \mathcal{P}^{\text{r},3}} \sum_{j \in \mathcal{P}^{\text{w},3}} (C_{i,t}^{\text{str-III}} + C_{j,t}^{\text{str-W-III}}) \\ &= \sum_g \left[\sum_{i \in \mathcal{U}_g^{\text{gm}}} c_{i,t}^{\text{m,r}}(s_{i,t}^{\text{r}}, a_{i,t}^{\text{r}}) + \sum_{j \in \mathcal{U}_g^{\text{gm}}} c_{j,t}^{\text{m,w}}(s_{j,t}^{\text{w}}, a_{j,t}^{\text{w}}) \right] \cdot \Delta \tau_g^{\text{gm}} \\ & - \sum_{i \in \mathcal{P}^{\text{r},3}} \sum_{j \in \mathcal{B}_i^{\text{r}} \cap \mathcal{P}^{\text{w},3}} \left[c_{i,j,t}^{\text{rp}}(a_{i,t}^{\text{r}}, a_{j,t}^{\text{w}}) + c_{i,j,t}^{\text{tc}}(a_{i,t}^{\text{r}}, a_{j,t}^{\text{w}}) \cdot \Delta T_{i,j}^{\text{m}} \right] \cdot L_{i,j}^{\text{co}} \\ & + \sum_g \left[\sum_{i \in \mathcal{U}_g^{\text{gm}}} \alpha_i F^{\text{r}}(s_{i,t}^{\text{r}}, a_{i,t}^{\text{r}}, (s_{i',t}^{\text{r}}, a_{i',t}^{\text{r}})_{i' \in \mathcal{I} \setminus \{i\}}, \mathbf{z}_t^{\text{r}}) \cdot \text{TF}_i \cdot \Delta \tau_g^{\text{gm}} \cdot c_{i,t}^{\text{ust,r}} \right. \\ & \left. + \sum_{j \in \mathcal{U}_g^{\text{gm}}} \alpha_j F^{\text{w}}(s_{j,t}^{\text{w}}, a_{j,t}^{\text{w}}, (s_{j',t}^{\text{w}}, a_{j',t}^{\text{w}})_{j' \in \mathcal{J} \setminus \{j\}}, \mathbf{z}_t^{\text{w}}) \cdot c_{j,t}^{\text{ust,w}} \cdot \Delta \tau_g^{\text{gm}} \right], \end{aligned} \quad (10)$$

where the first term represents the direct maintenance costs for Category-III roads and pipes, computed group by group using the aggregated construction durations. The second term captures cost reductions from joint maintenance, which depend on the actions applied to both roads and pipes. These efficiencies arise from eliminating redundant repaving and traffic control operations on co-located infrastructure. In particular, daily repaving costs, denoted as $c_{i,j,t}^{\text{rp}}(a_{i,t}^{\text{r}}, a_{j,t}^{\text{w}})$, account for expenditures on road surface restoration, including materials, machinery, and labor. Daily traffic control costs are denoted as $c_{i,j,t}^{\text{tc}}(a_{i,t}^{\text{r}}, a_{j,t}^{\text{w}})$, including costs for signage, equipment, vehicles, and personnel. The reduction in working duration, $\Delta T_{i,j}^{\text{m}}$, reflects the efficiency gains from joint maintenance of road i and pipe j , compared to conducting separate repairs on co-located components. These gains come from optimizing resource allocation, streamlining labor use, and avoiding redundant steps. $L_{i,j}^{\text{co}}$ represents the shared road area length between road i and pipe j . The third and fourth terms represent the indirect road user costs and water user costs arising from group maintenance, respectively. These terms are evaluated using the traffic and hydraulic simulation models described earlier, which capture the complex nonlinear interactions within the two networks.

2.5. Long-term cost modeling for co-located roads and pipes

Unlike short-term costs, which are mainly driven by temporary maintenance activities lasting days or weeks, long-term costs arise from

the gradual deterioration of roads and pipes over years. For a TI, these costs include increased traffic delays, accelerated vehicle depreciation, and higher fuel consumption. As road conditions worsen over time, such expenses escalate, creating a growing financial burden for users. Considering a road at spatial location i , the total long-term cost from year t to $t+1$ is given by:

$$C_{i,t}^{\text{lt,r}} = T^{\text{e}} \cdot L_i^{\text{r}} \cdot \text{TF}_i \cdot c_{i,t}^{\text{ult,r}}(s_{i,t}^{\text{int,r}}), \quad (11)$$

where $C_{i,t}^{\text{lt,r}}$ denotes the long-term user cost of road i from year t to $t+1$. The term T^{e} is the evaluation period, set as one year in this study. L_i^{r} is the length of road i , and TF_i is the annual traffic volume, obtained by converting AADT values into yearly totals. The coefficient $c_{i,t}^{\text{ult,r}}(s_{i,t}^{\text{int,r}})$ is a state-dependent annualized unit cost, which reflects the additional travel time, fuel consumption, and vehicle operating costs associated with degraded pavement conditions.

For WI, long-term costs arise from pipe deterioration through increased repair expenses, water loss, and infrastructure damage. As degradation progresses, frequent breaks lead to higher maintenance costs, service disruptions, and property damage, creating additional expenses for households and businesses from emergency repairs, higher utility bills, and interrupted service. Beyond these direct costs, co-location with roads introduces additional long-term costs through geographical dependencies. More frequent pipe breaks can cause repeated traffic disruptions, while the spatial relationship between roads and underlying pipes influences pipe failure rates. Studies show that pipe segments beneath intersections are more failure-prone than those parallel to roads or independent of them [42]. In practice, a single pipe may span multiple road segments with different topological relationships. To capture this variation, we partition pipes into smaller subsegments: Type I beneath intersections, subject to higher failure rates; Type II running parallel to roads, where failures directly disrupt traffic; and Type III independent of roads, whose failures impose no additional costs on the TI.

Fig. 3 shows a real-world example from our Ybor City testbed, illustrating a single pipe (ID: P441) and its topological relationships with co-located road segments. Type I subsegments, located beneath road intersections, exhibit the highest failure rates due to structural stress. Type II subsegments, aligned with roads but not crossing intersections, have lower failure rates than Type I but higher than Type III, which are spatially independent of roads. These types also differ in their impacts on the TI. Failures of Type II subsegments directly disrupt traffic on the co-located roads, causing temporary closures and delays. Failures of Type I subsegments at intersections can affect multiple roads simultaneously, leading to widespread disruptions. In contrast, failures of Type III subsegments, not adjacent to any roads, do not impose additional costs on the transportation network. To capture these differences, the total long-term cost of a pipe at location j from year t to $t+1$ is expressed as:

$$C_{j,t}^{\text{lt,w}} = \sum_{j \in \mathcal{J}} L_j^{\text{w}} \cdot c_j^{\text{leak}}(s_{j,t}^{\text{int,w}}) + \sum_{j \in \mathcal{J}} \sum_{k \in \mathcal{K}} \lambda_{j,k,t}(s_{j,t}^{\text{int,w}}) \cdot L_{j,k}^{\text{w}} \cdot c_{j,k}^{\text{break}}, \quad (12)$$

where the total long-term cost consists of two components: costs from water leaks due to pipe aging and expected costs from pipe breakages over the year. For any given pipe j , the leakage cost is calculated as the product of its length L_j^{w} and the annual leakage cost per meter c_j^{leak} . The unit leakage cost is state-dependent, with more deteriorated pipes incurring higher costs due to increased volume loss. To capture the long-term impact of pipe breakages, water pipes are partitioned into subsegments as described earlier. Let $k \in \mathcal{K}$ index the pipe sub-types, and let the failure rate of type- k subsegments at time t be given by Eq. (4) [40]. The expected number of failures for pipe j in year t is then $\lambda_{j,k,t}(s_{j,t}^{\text{int,w}}) \cdot L_{j,k}^{\text{w}}$, where $\lambda_{j,k,t}(s_{j,t}^{\text{int,w}})$ is the failure rate per unit length and $L_{j,k}^{\text{w}}$ is the length of type- k subsegments on pipe j . The average cost of a failure, denoted by $c_{j,k}^{\text{break}}$, accounts for reactive repair expenses, potential traffic delays from excavation, and penalties for unmet water demand.

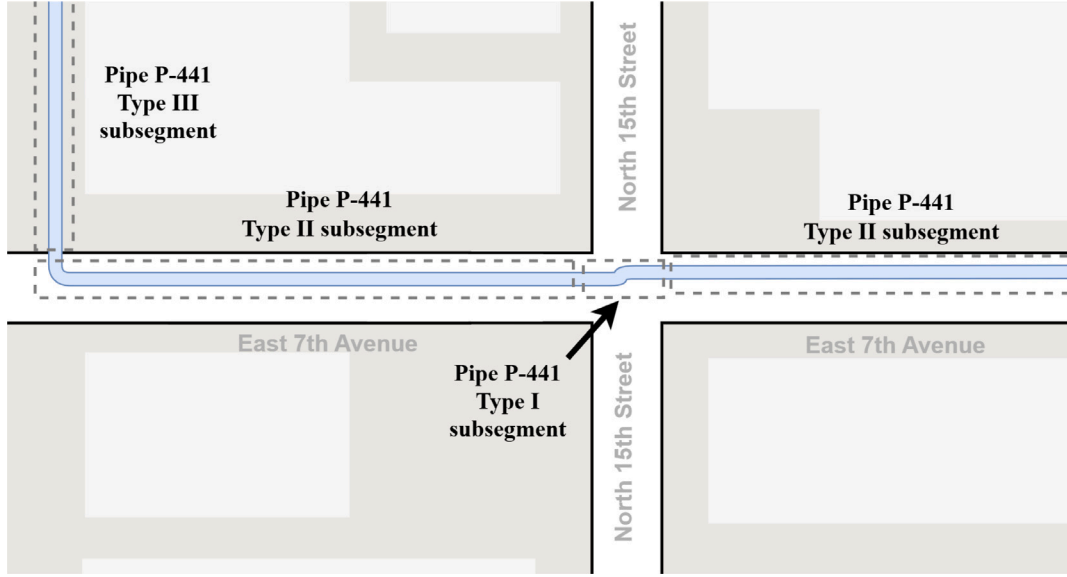


Fig. 3. Subsegments of pipes based on the topological relationship with roads.

2.6. Long-term maintenance optimization model for interdependent TI and WI

Building on the deterioration processes and cost components defined in Sections 2.4 and 2.5, we aggregate these into a system-wide, vectorized, and network-level framework. This formulation simultaneously captures intra-system dependencies through integrated traffic and hydraulic simulations, and inter-system interdependencies, which are encoded in the cost components. The resulting optimization model provides a joint maintenance policy that coordinates road and pipe decisions over the planning horizon to minimize the total expected cumulative cost of both short-term and long-term impacts.

Let the state vector $\mathbf{s}_t^r = (s_{1,t}^r, s_{2,t}^r, \dots, s_{I,t}^r)$ represent the deterioration conditions of a TI network with I road segments at the beginning of year t . The joint state space is defined as $S_{\text{joint}}^r = \{\mathbf{s}_t^r : s_{i,t}^r \in S^r\}$. Maintenance decisions at year t are collected into the joint action vector $\mathbf{a}_t^r = (a_{1,t}^r, a_{2,t}^r, \dots, a_{I,t}^r)$, with state-dependent action space $\mathcal{A}_{\text{joint}}^r(\mathbf{s}_t^r)$. The maintenance action space for each road segment is explicitly constrained based on its current deterioration state. Specifically, (i) if $s_{i,t}^r = 0$, only $a_{i,t}^r = 0$ (DN) is feasible; (ii) if $s_{i,t}^r = 1$, then $a_{i,t}^r \in \{0 \text{ (DN)}, 1 \text{ (MM)}\}$; and (iii) if $s_{i,t}^r = \Omega^r$, then $a_{i,t}^r = 2$ (PM) is mandatory. The transition probability matrix that characterizes the degradation uncertainty of a TI can be expressed as $\mathbf{P}^r(\mathbf{s}_{t+1}^r | \mathbf{s}_t^r, \mathbf{a}_t^r)$, which extends the component-level model in Section 2.2 to the network-wide state vector. While degradation dynamics and direct maintenance costs are segment-wise additive, user costs such as traffic delays depend on the joint state-action configuration of the entire TI network and therefore cannot be decomposed into independent segment-level terms.

Similar to the TI, for a WI consisting of J pipes, we define the state vector $\mathbf{s}_t^w = (s_{1,t}^w, s_{2,t}^w, \dots, s_{J,t}^w)$ to represent the deterioration states of all pipes at the beginning of year t . The joint state space is $S_{\text{joint}}^w = \{\mathbf{s}_t^w : s_{j,t}^w \in S^w\}$, where S^w denotes the individual state space of each pipe. Maintenance decisions are collected into the joint action vector $\mathbf{a}_t^w = (a_{1,t}^w, a_{2,t}^w, \dots, a_{J,t}^w)$, with feasible choices determined by state-dependent constraints: (i) if $s_{j,t}^w = 0$, only $a_{j,t}^w = 0$ (DN) is feasible; (ii) if $s_{j,t}^w \leq 9$, then $a_{j,t}^w \in \{0 \text{ (DN)}, 1 \text{ (MM)}\}$; and (iii) if $s_{j,t}^w = \Omega^w$, then $a_{j,t}^w = 2$ (PM) is mandatory. These rules align maintenance options with each pipe's current condition and ensure consistency across the WI network.

Considering a H -year planning horizon with a time index set $\mathcal{T} = \{1, 2, \dots, H\}$, the proposed network-wide optimization model for joint long-term maintenance planning of co-located TI and WI systems can

be formulated as:

$$\min_{\pi} J_1^{\pi}(\mathbf{s}, \mathbf{a}) \quad (13)$$

$$J_t^{\pi}(\mathbf{s}, \mathbf{a}) = \underbrace{C_t(\mathbf{s}, \mathbf{a})}_{\text{Stage cost}} + \underbrace{\gamma \mathbb{E}_{\mathbf{a}' \sim \pi(\cdot | \mathbf{s}), \mathbf{s}' \sim p(\cdot | \mathbf{s}, \mathbf{a})} [J_{t+1}^{\pi}(\mathbf{s}', \mathbf{a}')] }_{\text{Expected future cost}} \quad \text{for } t = 1, \dots, H-1,$$

$$J_H^{\pi}(\mathbf{s}, \mathbf{a}) = \underbrace{C_H(\mathbf{s}, \mathbf{a})}_{\text{Terminal cost}}$$

where

$$\begin{aligned} C_t(\mathbf{s}, \mathbf{a}) := & \sum_{i \in \mathcal{P}^{r,1}} C_{i,t}^{\text{st},r-1}(\mathbf{s}_i^r, \mathbf{a}_i^r) + \sum_{j \in \mathcal{P}^{w,1}} C_{j,t}^{\text{st},w-1}(\mathbf{s}_j^w, \mathbf{a}_j^w) \\ & + \sum_{j \in \mathcal{P}^{w,2}} C_{j,t}^{\text{st},w-\text{II}}(\mathbf{s}, \mathbf{a}) + \sum_{i \in \mathcal{P}^{r,3}} \sum_{j \in \mathcal{P}^{w,3}} (C_{i,t}^{\text{st},r-\text{III}}(\mathbf{s}_i^r, \mathbf{a}_i^r) + C_{j,t}^{\text{st},w-\text{III}}(\mathbf{s}_j^w, \mathbf{a}_j^w)) \\ & + \sum_{i \in \mathcal{P}^r} \sum_{j \in \mathcal{P}^w} (C_{i,t}^{\text{lt},r}(\mathbf{s}_i^r, \mathbf{a}_i^r) + C_{j,t}^{\text{lt},w}(\mathbf{s}_j^w, \mathbf{a}_j^w)) \quad \text{for all } (\mathbf{s}, \mathbf{a}), (\mathbf{s}', \mathbf{a}') \in S \times \mathcal{A}. \end{aligned}$$

$J_t^{\pi}(\mathbf{s}, \mathbf{a})$ represents the total expected cumulative cost from decision epoch t to the end of the planning horizon ($t = H$) under policy π , which maps each state $\mathbf{s}$ to an action $\mathbf{a}$. The state vector $\mathbf{s} = (s_t^r, s_t^w)$ captures the deterioration conditions of all roads and pipes at epoch t , and the joint action vector $\mathbf{a} = (a_t^r, a_t^w)$ specifies the corresponding maintenance decisions. The joint state space S and action space $\mathcal{A}$ are the Cartesian products of all component-level state and action spaces across the two networks. The objective in Eq. (13) is to minimize the total expected cumulative cost $J_1^{\pi}(\mathbf{s}, \mathbf{a})$ from the initial epoch ($t = 1$) to the end of the horizon ($t = H$) by selecting an optimal policy π .

The expected cumulative cost at decision epoch t can be expressed recursively as the sum of two components: the immediate cost $C_t(\mathbf{s}, \mathbf{a})$ and the future expected cost from epoch $t+1$ to the end of the horizon, $\gamma \mathbb{E}_{\mathbf{a}' \sim \pi(\cdot | \mathbf{s}), \mathbf{s}' \sim p(\cdot | \mathbf{s}, \mathbf{a})} [J_{t+1}^{\pi}(\mathbf{s}', \mathbf{a}')]$. The immediate cost $C_t(\mathbf{s}, \mathbf{a})$ aggregates direct maintenance, short-term user, and long-term user costs across all road and pipe components at epoch t , categorized as in Section 2. The future expected cost accounts for deterioration uncertainty, modeled by the transition probability $p(\mathbf{s}' | \mathbf{s}, \mathbf{a})$, and depends on the next action $\mathbf{a}'$ chosen under policy $\pi(\cdot | \mathbf{s}')$. The discount factor $\gamma \in (0, 1)$ balances immediate and future costs: values close to 1 emphasize long-term effects, while smaller values emphasize short-term outcomes. At the terminal epoch H , the terminal cost $C_H(\mathbf{s}, \mathbf{a})$ captures the final impacts of maintenance actions without further recursion.

Given the high dimensionality of the joint state-action space and the nonlinear interdependencies captured through traffic and hydraulic simulations, traditional optimization approaches become impractical. We therefore adopt a DRL framework, which efficiently handles scalability while directly incorporating nonlinearities by interacting with the simulation environment. Specifically, we employ the Deep Q-Network (DQN) algorithm to approximate the optimal state-action cost-to-go function, denoted as $J^{\pi^*}(s, a)$. In this formulation, the stage-wise cost $C_t(s, a)$ serves as the immediate feedback signal, and the cumulative cost is recursively estimated using the Bellman equation. The DQN model is trained by minimizing the temporal difference (TD) error:

$$\delta = -C_t(s, a) + \gamma \min_{a'} \hat{J}(s', a'; \theta^-) - J(s, a; \theta), \quad (14)$$

where θ and θ^- denote the parameters of the current network J and the target network $\hat{J}$, respectively. Both networks are implemented as deep neural networks used to approximate the state-action cost-to-go function. The current network is updated frequently to fit new observations, while the target network is updated more slowly to provide a stable reference for learning. The network is trained by minimizing the expected squared TD error. The overall DQN procedure is summarized in Algorithm 1, which outlines the use of the experience replay buffer, target network updates, and cost-to-go function training steps to iteratively improve the maintenance policy.

Algorithm 1 DQN for Maintenance Planning of Co-located Roads and Pipes

```

1: Initialize an empty replay buffer  $B$ 
2: Initialize cost-to-go network  $J(s, a; \theta)$  with random weights  $\theta$ 
3: Initialize target network  $\hat{J}(s, a; \theta^-)$  by setting  $\theta^- \leftarrow \theta$ 
4: for episode index  $m = 1$  to  $M$  do
5:   Initialize initial state  $s = (s_1^r, s_1^w)$  from environment
6:   for time step  $t = 1$  to  $H$  do
7:     if random number  $< \epsilon$  then
8:       Select a random action  $a$ 
9:     else
10:      Select action that minimizes  $J(s, a; \theta)$ 
11:    end if
12:    Simulate environment to obtain cost  $C_t(s, a)$  and next state
       $s' = (s_{t+1}^r, s_{t+1}^w)$ 
13:    Store transition  $(s, a, C_t, s')$  in  $B$ 
14:    Sample mini-batch of transitions from  $B$ 
15:    Compute TD target:
16:       $y = \begin{cases} -C & \text{if } s' \text{ is terminal} \\ -C + \gamma \min_{a'} \hat{J}(s', a'; \theta^-) & \text{otherwise} \end{cases}$ 
17:    Update  $\theta$  by minimizing squared loss:
18:       $\mathcal{L}(\theta) = (y - J(s, a; \theta))^2$ 
19:    Every  $N$  steps, update target network using soft update:
20:       $\theta^- \leftarrow \eta \theta + (1 - \eta) \theta^-$ 
21:   end for
22: end for
```

Hyperparameters (fixed across experiments). Replay buffer $|B| = 10^6$ with uniform random sampling (FIFO eviction); minibatch size $B = 128$; learning rate 1×10^{-4} (Adam); discount $\gamma = 0.95$; exploration ϵ -greedy with $\epsilon_0 = 1.0 \rightarrow \epsilon_{\min} = 0.01$ (decay 0.992 per step); soft target update $\theta^- \leftarrow \eta \theta + (1 - \eta) \theta^-$ with $\eta = 0.005$, every 4 steps; reward normalization $\text{sign}(x) \log_{10}(1 + |x|)$ applied to $x = -C_t$.

3. Case study

3.1. Overview

To evaluate the effectiveness of the proposed strategy, we benchmark it against two alternatives (S1: reactive and S2: proactive) using a real-world case study of Ybor City, Tampa, FL. The goal is to demonstrate the benefits of joint, interdependency-aware maintenance for

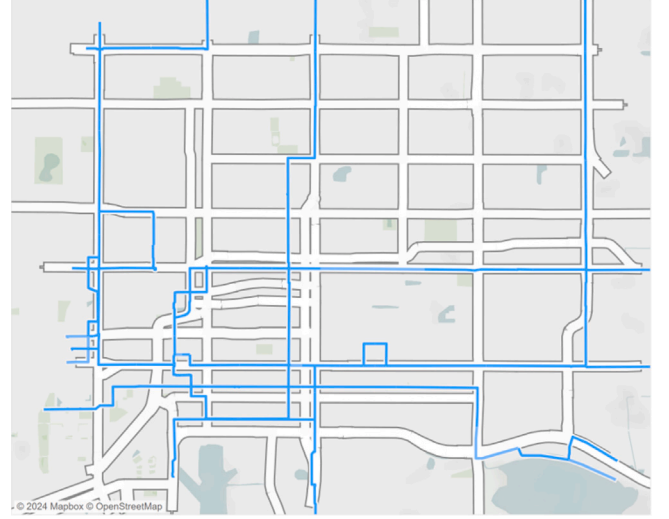


Fig. 4. The co-located TI and WI in Ybor City, Tampa FL.

co-located infrastructures. S1 triggers PM only after failure, while S2 takes preventative actions to avoid costly repairs but treats roads and pipes independently, ignoring spatial co-location. As a result, TI-WI interdependencies and variations in pipe failure rates due to road topology are not considered. In contrast, the proposed strategy is proactive and explicitly accounts for TI-WI co-location, leveraging joint maintenance to optimize costs through interdependency-aware planning.

3.2. Problem and computational experiment setup

To evaluate the approaches in a realistic setting, we developed a testbed emulating the co-located road and water pipe networks in Ybor City, Tampa, FL. The testbed replicates local conditions, consisting of asphalt-paved roads and underground PVC water pipes [47]. Ybor City spans 2.6 km² with 4821 residents and was selected for its historical significance as one of Tampa's oldest districts [48]. The area faces challenges due to its aging infrastructure, including both TI and WI, highlighting the need for proactive maintenance and making it ideal for evaluating the proposed framework.

The interconnected TI and WI networks for the testbed are shown in Fig. 4, with blue lines for pipes and white grids for roads. Road data was sourced from OpenStreetMap (OSM) [49] and processed via `osmWebWizard.py` [50]. Water pipe data was obtained from Abdel-Mottaleb et al. [45]. Road and pipe networks were modeled at the segment level to define the spatial resolution of the testbed. Specifically, road segments were defined as the sections between two intersections, inclusive of lanes in both directions and turning lanes at intersections where applicable, and pipe segments were defined as the portions between two pipe joints. This segmentation resulted in 179 road segments and 95 pipe segments. To assess TI disruptions (e.g., from maintenance or failures), we incorporated real traffic data, such as Annual Average Daily Traffic (AADT) records [51] into our testbed. In addition, the SUMO traffic simulator was employed to capture practical traffic constraints based on the OSM network, including road geometry, lane configurations, speed limits, intersection control (signals and right-of-way), vehicle dynamics (e.g., acceleration, car-following, and lane-changing), and driver behavior. Maintenance-related closures reduce the effective capacity of affected links, while SUMO enforces feasible detours and queuing dynamics at intersections. To evaluate the demand shortfall of WI due to pipe maintenance, simulation outputs from a hydraulic model [45,46] were integrated.

Table 2
Discretized states and cut points for roads and pipes.

$s_{i,t}^r$	IRI value (m/km)	$s_{j,t}^w$	Age (yr.)	Condition
0	$IRI \leq 0.95$	0 - 19	$0 \leq \text{Age} \leq 19$	Very Good
1	$0.95 < IRI \leq 1.50$	20 - 39	$20 \leq \text{Age} \leq 39$	Good
2	$1.50 < IRI \leq 1.90$	40 - 69	$40 \leq \text{Age} \leq 69$	Fair
3	$1.90 < IRI \leq 2.84$	70 - 98	$70 \leq \text{Age} \leq 98$	Poor
4	$IRI > 2.84$	99	$\text{Age} > 98$	Failure

Table 3
Cost information summary of the case study (\$: US Dollar, m: meter).

Type of cost	Condition states				
Cost settings of road	State 0	State 1	State 2	State 3	State 4
$c_i^{m,r}$ for $a_i^r = \text{MM}$ (\$/m)	–	131	131	131	131
$c_i^{m,r}$ for $a_i^r = \text{PM}$ (\$/m)	–	627	958	1,336	1,804
c_i^{tc} (\$/m/day)	20	20	20	20	20
c_i^{rp} (\$/m)	58	58	58	58	58
$c_i^{ult,r}$ (\$/veh/m)	0.0166	0.0233	0.0332	0.0499	0.0831
$c_i^{ust,r}$ (\$/veh/h)	18.27	18.27	18.27	18.27	18.27
Cost settings of pipe	State 0–19	State 20–39	State 40–69	State 70–98	State 99
$c_i^{m,w}$ for $a_i^w = \text{MM}$ (\$/m)	–	119	119	119	119
$c_i^{m,w}$ for $a_i^w = \text{PM}$ (\$/m)	–	777	777	777	777
c_i^{leak} (\$/m)	–	60	119	239	3,883
c_i^{break} (\$/m)	656	656	656	656	656
$c_i^{ust,w}$ (\$/gal)	–	0.15	0.15	0.15	0.15

Note: Cost parameters are mainly adapted from existing work [23]. Units differ across cost categories by design. Direct maintenance, repaving, leakage, and break costs are expressed per meter (\$/m). Traffic control costs are per meter per day (\$/m/day) to reflect both unit length and maintenance duration. User delay costs are short-term in \$/veh/h (simulation outputs traffic delay per vehicle converted to hours and scaled by AADT and intervention duration) and long-term in \$/veh/m (annualized delay per vehicle per unit length). Water service shortfall costs are in \$/gal, reflecting unmet demand.

In this study, the states of roads and water pipes were represented using discrete levels. The IRI was selected as the performance indicator for road deterioration [52]. To enhance interpretability, the IRI was discretized into five condition states, labeled 0 to 4, representing conditions ranging from “Very Good” to “Failure”, as detailed in Table 2. A failure threshold of 2.84 m/km was established for road segments. When a road segment reaches the failure state ($IRI > 2.84$ m/km), a PM action is triggered, restoring the road to state 0. The discrete road states were derived using eight years of historical data from FDOT and estimated using the Maximum Likelihood Estimation (MLE) method [23]. For water pipes, states were determined by their ages since installation, ranging from 0 to 99, with 99 representing the failure state. Maintenance actions can extend the lifespan of pipes, so the actual service life of the pipes in this study may exceed 100 years, reflecting real-world conditions where properly installed PVC pipes typically have a lifespan beyond a century [53]. Similar to the roads, when a pipe reaches the failure state, a PM action, such as replacement, is undertaken. Cost parameters in Eq. (13) were based on [23,54–56] (Table 3). In this study, the cost data are U.S.-specific. When applying the model to other regions, region-specific data should be incorporated and the maintenance model recalibrated to ensure appropriate transferability. The overall planning horizon was set as $H = 10$. In Ybor City, 179 road and 95 pipe segments were considered. For each segment, the applicable actions are DN, MM, and PM. At any year t , the state and action spaces are 5^{179} and 3^{95} .

The environment of Ybor City encompasses various types of pipe segments, each exhibiting a distinct spatial relationship to the roads. These spatial configurations include scenarios where the pipe runs directly beneath and intersects the road at a junction (Type I), runs parallel beneath the road (Type II), or is located independently of the road network (Type III). An example is illustrated in Fig. 3. As shown in Eq. (4), each configuration experiences different stress levels from traffic loads and vibrations, which ultimately influence their failure rates [42]. In the case study, for PVC pipe segments, we set the

coefficient of growth of the breakage rate, A , equal to 0.056 based on the work of Ahopelto and Vahala [57]. We set the initial failure rates ($\lambda_{k,i0}$) to 0.018 (Type I), 0.01 (Type II), and 0.005 (Type III) based on existing empirical studies [42,57], given the limited availability of detailed local failure data. In future, these parameters should be calibrated using detailed local pipe failure records when available. Network-wide traffic delay arises from both maintenance actions and road deterioration, with associated short-term and long-term user costs detailed in Section 2. Daily delays are simulated and then aggregated to annual totals. To estimate the short-term impacts of maintenance actions, we develop a surrogate model to approximate traffic delays using SUMO-simulated samples under various road blockage scenarios. Directly solving the optimization problem would otherwise require numerous interactions with the microsimulation, which is computationally expensive. To mitigate potential surrogate errors, we construct a comprehensive training dataset of 5278 samples that capture diverse blockage patterns, including multiple simultaneous closures and variations in road features such as traffic speed, hourly flow, adjacent roads, lane count, length, and functional classification. We then compare several machine learning methods, e.g., linear regression, random forest, gradient boosting, and neural networks, and identify XGBoost as the most accurate. The resulting surrogate model achieves an average relative squared error (RSE) of 0.055 and a mean squared error (MSE) of 0.1587 across different blockage scenarios. The predicted values represent seconds of delay per vehicle traveling across the network.

As stated in Section 2.6, to solve the problem, we employ the DQN algorithm. The computational complexity of DQN is primarily determined by (i) the size of the neural network architecture and (ii) the time required to interact with the environment through sampling and gradient updates [58]. In our problem setting, a three-layer multilayer perceptron with hidden dimensions [256, 256, 128] and ReLU activations was sufficient to capture the decision-making patterns. Training converged within 3.4 h under our experimental setup, which used an Intel Core i7 CPU (8 cores, 16 threads, 3.6 GHz), 32 GB RAM, and an

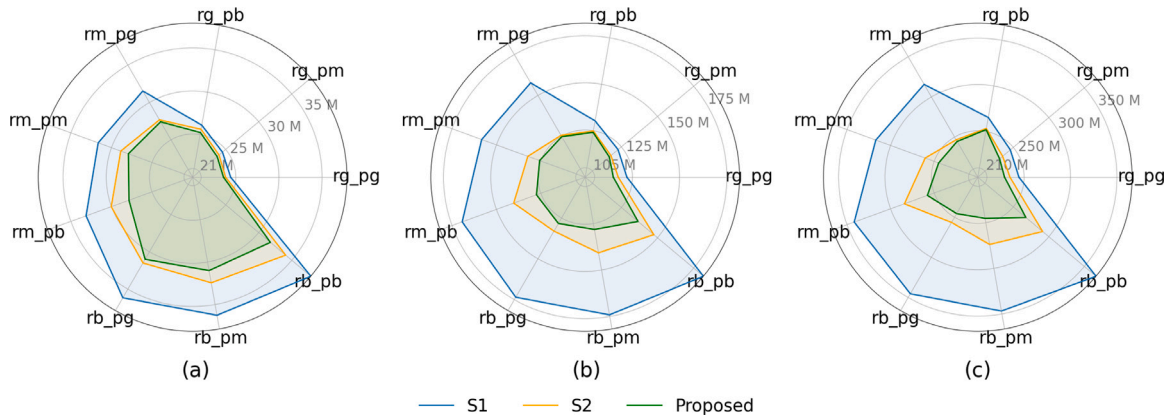


Fig. 5. The total expected cumulative cost comparisons among the proposed (in green), S1 (in blue) and S2 (in orange) strategies over 1, 5 and 10 years.

NVIDIA GeForce RTX 3080 GPU (8 GB VRAM), with Python 3.9.12 and PyTorch 2.0.0+cu117 on Windows 10. The use of an experience replay buffer with a capacity of 10^6 further improved efficiency by allowing the DQN to reuse past interactions without repeatedly querying the environment models. DQN was chosen over more advanced reinforcement learning methods, such as Proximal Policy Optimization (PPO), primarily for its computational efficiency and comparability across different maintenance strategies. In our experiments, both DQN and PPO achieve comparable convergence results and their reward functions tended to stabilize after approximately 3000 to 4000 episodes, indicating similar performance from the perspective of convergence behavior. However, the average per-episode computation time differed significantly: about 2.4 s for DQN versus 29.1 s for PPO. This observation is consistent with the existing literature, which shows that PPO generally requires more computation per iteration due to repeated updates of both policy and value networks with clipped surrogate objectives, whereas DQN only updates a single value network using replayed minibatches [59].

3.3. Performance comparison

In this section, we compare the performance of the proposed strategy against two conventional benchmarks that neglect TI–WI dependencies. The first benchmark, strategy S1, is a reactive, rule-based strategy. Under this strategy, perfect maintenance (PM) is only performed after component failures occur. It performs separate maintenance for TI and WI, and therefore overlooks the geographical dependencies and potential cost savings from joint maintenance. In contrast, both strategy S2 and our proposed approach are proactive in nature, but they differ significantly in how dependencies are addressed. Strategy S2 performs separate maintenance planning for TI and WI, failing to consider how failures in one system may disrupt the other. For instance, S2 does not account for the additional traffic disruption caused by pipe breaks or the influence of heavy traffic loads on the deterioration of co-located pipes. It also overlooks cost-saving opportunities from joint operations, such as sharing repaving and traffic control when both roads and pipes in the same location require repair. Our proposed method, however, explicitly accounts for these dependencies and coordinates maintenance across systems to capture both geographic dependencies and opportunities for cost savings.

To comprehensively evaluate the impact of initial degradation states on cost performance, we define three overall degradation scenarios for both TI and WI—representing good, medium, and poor conditions. For TI: (i) in the “bad” scenario, 75% of road segments are in state 3 and the rest in state 4; (ii) in the “medium” scenario, all roads are in state 2; and (iii) in the “good” scenario, roads are randomly initialized to state 0 or 1 with equal probability. For WI: (i) the “bad” scenario samples pipe

ages uniformly between 70 and 99 years; (ii) the “medium” scenario samples between 40 and 69 years; and (iii) the “good” scenario samples between 0 and 39 years. These definitions result in nine combinations of scenarios. We denote them using abbreviations such as rg_pb and rb_pm, where “r” and “p” refer to road and pipe conditions, and “g”, “m”, and “b” denote good, medium, and bad scenarios, respectively.

Fig. 5 compares cumulative costs of the proposed strategy versus S1 and S2 over 1, 5, and 10 years. In all three timeframes, the proposed strategy consistently yields the lowest cost across all nine scenarios. This highlights the benefits of proactive maintenance that anticipates future degradation and considers TI–WI interdependencies. In contrast, the reactive rule-based strategy S1, which neither anticipates deterioration nor considers TI–WI interdependencies, incurs the highest cumulative costs. The proactive strategy S2 generally performs better than S1 but worse than the proposed method. While S2 emphasizes the value of preemptive maintenance by anticipating deterioration, it fails to consider the physical and operational interdependencies between roads and pipes. Cost-saving benefits of the proposed strategy are most significant when the TI is in a “bad” condition scenario, or when both TI and WI are in “medium” or “bad” states. TI plays a relatively more dominating role in cost savings, mainly because roads constitute a larger share of the infrastructure network in Ybor City, with more mileage and higher user and maintenance costs. Fig. 6 complements Fig. 5 by reporting grouped bars of mean cumulative costs with 95% bootstrap confidence intervals across the nine scenarios and the 1-, 5-, and 10-year horizons. The proposed strategy (green) consistently remains below S2 (orange) and S1 (blue), demonstrating statistically robust cost advantages that persist under spatial variations in initial deteriorating conditions and other scenario-specific variations.

To better understand why the proposed strategy achieves greater cost savings compared to S1 and S2, we decompose 5-year overall costs into short-term and long-term components. Short-term costs arise from maintenance-induced disruptions to roads and pipes, while long-term costs capture user impacts due to infrastructure deterioration and failures. As shown in Fig. 7(a), both the proposed strategy and S2, which adopt proactive maintenance, result in higher short-term costs than the reactive strategy S1. These higher costs stem from more preventative actions aimed at delaying deterioration and avoiding future failures. Notably, the proposed strategy tends to take even more preventative actions than S2, as it accounts not only for individual deterioration in TI and WI, but also for their interdependencies and combined consequences on long-term user costs. In return, both proactive strategies significantly reduce long-term costs compared to S1, as shown in Fig. 7(b). These long-term costs consist of traffic delays and vehicle depreciation for road users caused by road deterioration, as well as annual leakage and breakage costs resulting from pipe deterioration.

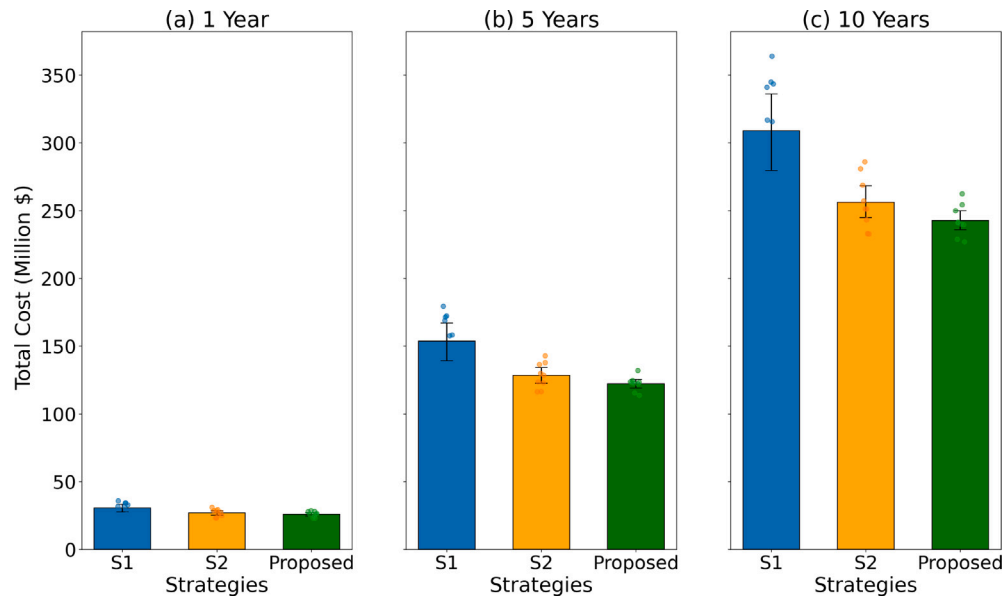


Fig. 6. Cost comparison among different maintenance strategies, considering the variability of initial deterioration conditions of roads and pipes.

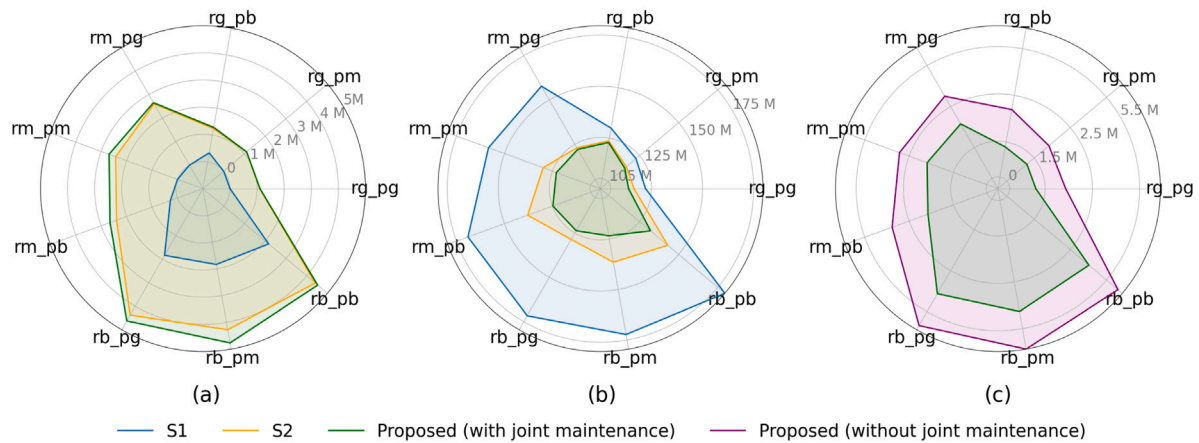


Fig. 7. Decomposed cost components comparison of (a) total short-term costs and (b) total long-term costs under 5-year horizon, and (c) cost-saving benefits of joint maintenance.

To illustrate, we examine the rm_pb scenario and compare its short-term and long-term costs under each strategy. Under the same initial degradation states, S1 performs PM on 5 pipes and no roads, resulting in a short-term cost of \$0.28M and a long-term cost of \$169.02M over five years. S2 takes more preventative actions, with PM on 5 roads and 21 pipes, and MM on 62 roads and 33 pipes, leading to a short-term cost of \$2.39M and a long-term cost of \$137.88M. The proposed strategy takes the most extensive actions, with PM on 21 roads and 25 pipes, and MM on 65 roads and 38 pipes, resulting in a short-term cost of \$2.65M and a long-term cost of \$124.78M. Both proactive strategies incur higher short-term costs but yield substantial long-term savings. This reflects the fundamental trade-off in our framework, which aims to balance short-term disruptions with long-term deterioration costs to minimize the overall cumulative cost of co-located TI and WI systems. Moreover, the proposed strategy's slightly higher short-term cost compared to S2 leads to greater reductions in long-term user costs, highlighting the benefits of accounting for interdependencies in joint maintenance planning. The first two panels in Fig. 8 summarize

mean short- and long-term costs with 95% nonparametric bootstrap confidence intervals, confirming the trade-off seen in Fig. 7: proactive strategies raise short-term spending but deliver larger long-run savings than S1. Besides, the third panel shows that, for the same repair decisions, coordinated joint maintenance (proposed) reduces repaving and traffic-control expenditures relative to S2's separate actions, improving cost efficiency. These findings are statistically robust across scenario-specific variations and various initial deteriorating conditions.

The proposed strategy takes more proactive actions than S2 due to two main factors. First, it anticipates user costs arising from TI-WI interdependencies, such as road disruptions caused by pipe failures and accelerated pipe deterioration from traffic loads. As a result, it performs more repairs, especially when either system is in poor condition. Second, by implementing joint maintenance for co-located roads and pipes, the proposed strategy reduces repaving and traffic control costs compared to S2's separate actions. This coordination boosts cost efficiency, allowing more maintenance under the same budget. Fig. 7(c) shows that under the same repair decisions, joint maintenance in

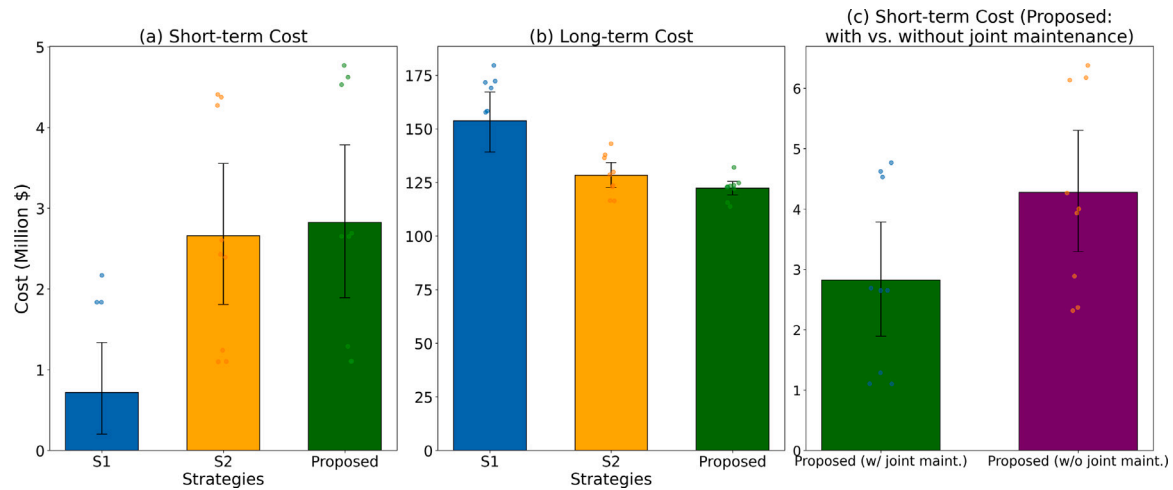


Fig. 8. Short- and long-term cost comparison among maintenance strategies over 5 years, considering the variability of initial deterioration conditions of roads and pipes. The figure comprises three panels: (a) short-term costs; (b) long-term user costs; and (c) joint vs. separate (without joint) maintenance under identical repair decisions in the proposed strategy.

the proposed strategy is more cost-effective and allows more proactive interventions than separate maintenance.

3.4. Comparisons of maintenance actions

Fig. 9 compares the total number of maintenance actions (DN, MM, PM) for TI and WI across nine degradation scenarios. The stacked bars show aggregated actions, with roads displayed above and pipes below. Under the reactive strategy S1, DN dominates for both roads and pipes, with PM applied only once a segment reaches a degradation threshold. In contrast, S2 and the proposed strategy proactively use both MM and PM in all scenarios, and the share of PM increases as conditions worsen. Although this raises short-term costs, it substantially reduces long-term and total costs. Compared with S1, the proposed strategy undertakes more proactive actions overall, applying more MM and PM to both roads and pipes in every scenario. This reflects its consideration of interdependencies, including disruptions from pipe failures, faster pipe deterioration beneath roads, and cost savings from coordinated maintenance.

In addition to the above differences, the top panels of Fig. 9 reveal two further insights. First, as road conditions worsen from medium to bad, the proposed strategy adopts substantially more PM actions than S2. For example, in the top left subplot (dashed box), where WI is fixed (e.g., pb), the proposed strategy performs 21 PM actions under medium TI (rm) compared with 5 by S2; as TI deteriorates to rb, these numbers increase to 53 and 38, respectively. This reflects the proposed strategy's greater emphasis on mitigating long-term user costs from poor roads. Second, as pipe conditions decline, the proposed strategy increases road maintenance even when road conditions remain unchanged. For instance, when TI is medium (rm), the proposed strategy performs 21 PM and 65 MM road actions (top left subplot), compared with 10 PM and 59 MM in the top middle subplot. In contrast, S2 applies nearly identical road maintenance regardless of pipe conditions. This illustrates the proposed strategy's recognition of road-pipe interdependence: deteriorating pipes justify more proactive road work. Since pipe repairs inevitably cause short-term user costs on co-located roads (traffic delays, repaving, traffic control), the proposed strategy aligns road and pipe interventions so that both works occur during the same disruption window, thereby minimizing repeated closures and yielding cost savings. Unlike road maintenance, which is sensitive to pipe conditions, pipe actions (bottom panels of Fig. 9) vary little across

road scenarios. This occurs for two reasons. First, road work rarely affects underground pipes directly, so pipe performance and user costs remain largely unchanged. Second, road-to-pipe dependency is mostly static: deteriorated roads increase pipe failures beneath them due to loads and vibrations, but this effect stems from spatial configuration rather than dynamic road condition changes.

3.5. Impact of various contributing factors on maintenance actions

Besides degradation performance, we examine additional factors influencing the proposed strategy's maintenance actions. Under a fixed scenario (rm_pb), three key TI maintenance factors are traffic volume, co-located pipe count, and road length. Fig. 10 shows the marginal effects of these factors on road maintenance. Subplot (a) shows traffic volume significantly influences maintenance actions. Among roads with very heavy traffic volumes (AADT > 52,800), 28% received PM and 32% received MM. In contrast, among segments with light traffic volumes (AADT ≤ 4800), none received PM, and only 27% received MM. Moreover, only 41% of roads with very heavy traffic received no maintenance, compared to 73% of light-traffic roads. These results indicate that the proposed strategy prioritizes PM for high-traffic roads. This is because delayed repairs on high-traffic roads lead to higher long-term costs like delays and vehicle wear. Moreover, without proactive maintenance, the total failure in high-traffic roads can cause major short-term disruptions and longer repairs. These costs can be mitigated through earlier preventive actions. Similar patterns are observed in subplots (b) and (c). Roads with more co-located pipes or greater length contribute more to long-term user costs. Thus, the strategy favors MM or PM for these segments to lower future costs.

Similarly, under the fixed rm_pb scenario, we identified three factors influencing pipe maintenance: pipe length, co-located road count, and pipe type. Because a pipe may include subsegments of different types, we classify it by the highest-risk subsegment: if any Type I is present, the pipe is Type I; if not, but at least one Type II is present, it is Type II; otherwise, it is Type III. This classification reflects differences in failure risk and pipe-road interdependencies. As shown in Fig. 11(a), the proposed method prioritizes longer pipes, which are more critical to the network and pose higher risks of costly repairs, extended service interruptions, and leakage. Fig. 11(b) further shows that pipes with more co-located roads are prioritized because they can trigger cascading disruptions and raise user costs for both networks. These

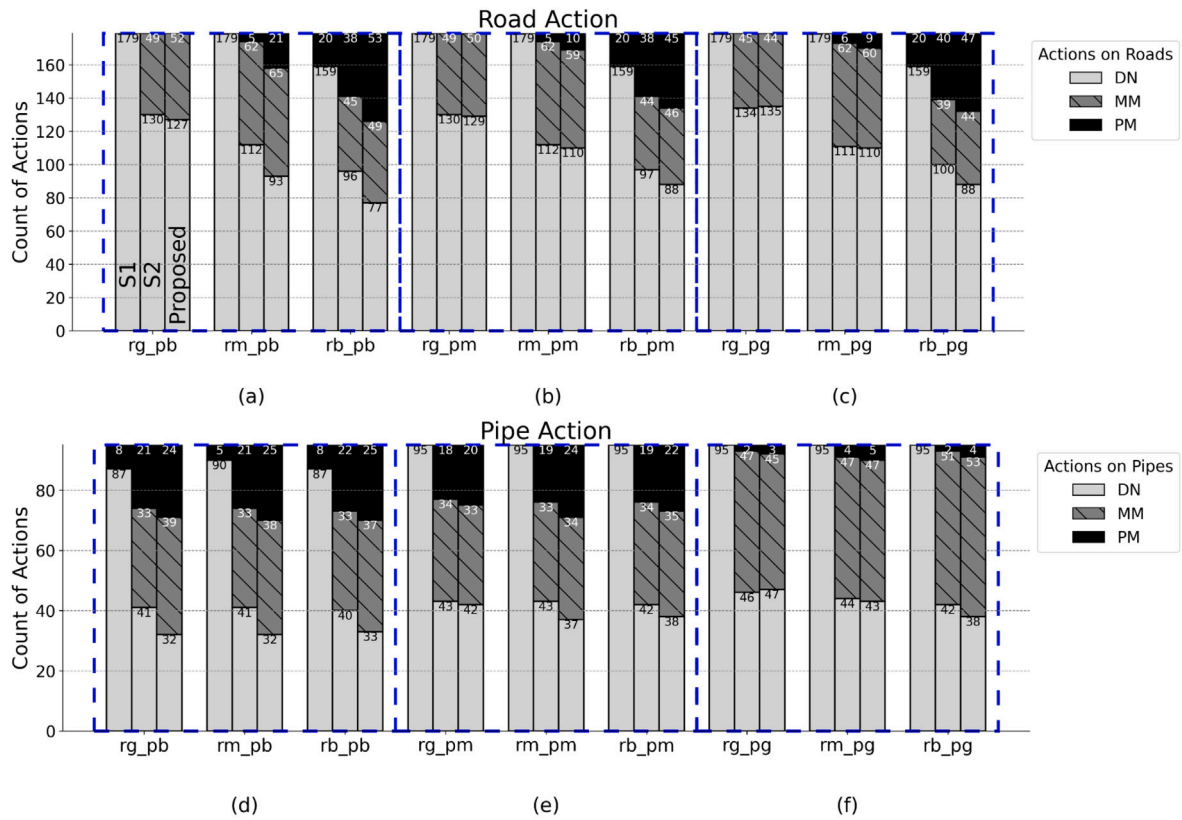


Fig. 9. Aggregated actions counts cross different strategies and scenarios.

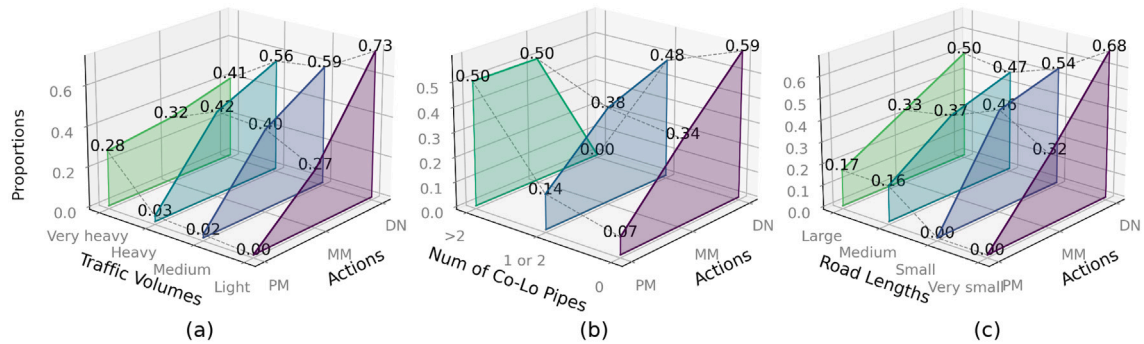


Fig. 10. Impact of contributing factors in road maintenance actions.

pipes also tend to contain Type I or II subsegments with higher failure rates, requiring proactive actions (Fig. 11(c)). Addressing such high-risk pipes is therefore essential to mitigate severe long-term costs from repairs and service disruptions.

3.6. Spatial heterogeneity of maintenance actions

This subsection conducts a sensitivity analysis to examine how the spatial distribution of initial deterioration affects maintenance patterns under the proposed strategy. With other factors held constant, Fig. 12 compares maintenance actions under two initial-state scenarios: subplot (a) assumes uniform deterioration, while subplot (b) reflects

the real-world distribution in Ybor City. PM, MM, and DN actions are shown as red, gray, and green lines, respectively. In the uniform case, maintenance actions are relatively balanced across the network, with local variations reflecting component features and traffic volume noted in Section 3.5. By contrast, in Ybor City the southwest region, including the Historic District, exhibits worse initial conditions due to aging infrastructure and is designated as part of the Ybor City Redevelopment Area targeting infrastructure improvements [60]. Consequently, maintenance actions concentrate in this region, showing that spatially uneven initial deterioration is more decisive in shaping maintenance allocations than individual component differences. The proposed strategy responds to these conditions by prioritizing high-need areas.

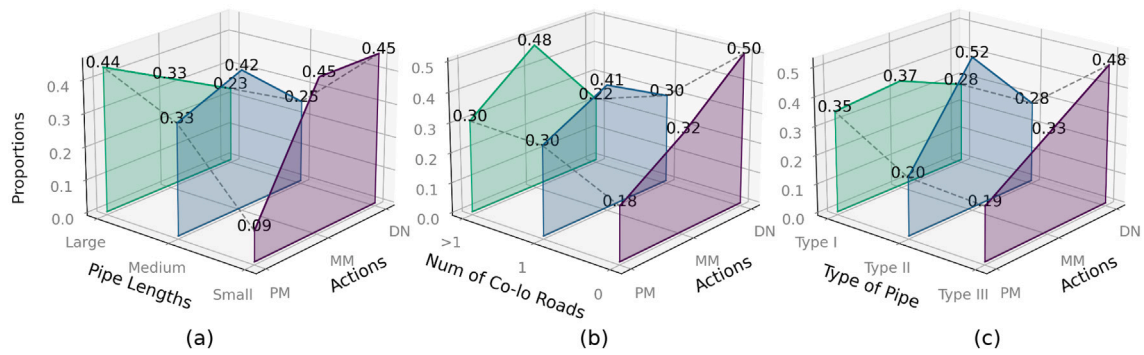


Fig. 11. Impact of contributing factors in pipe maintenance actions.

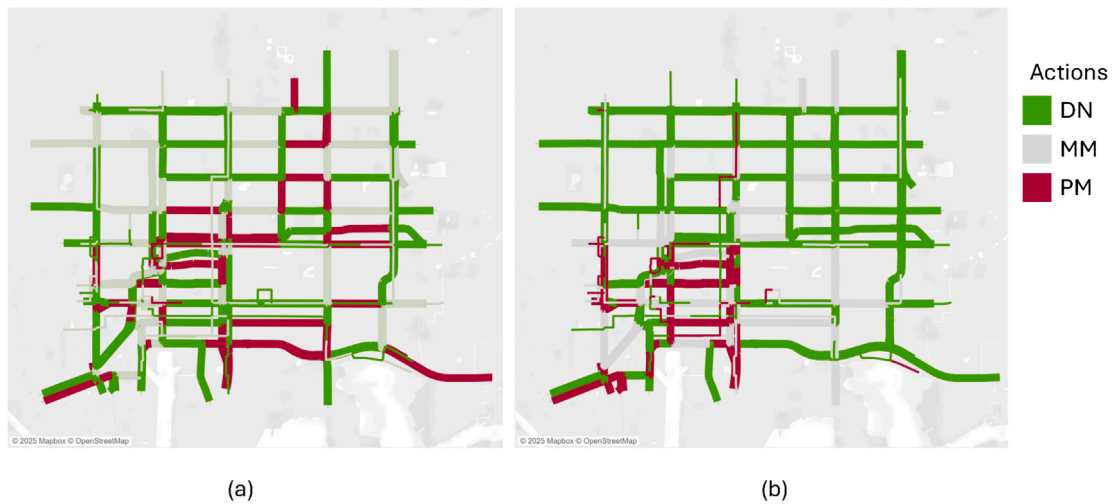


Fig. 12. Spatial pattern of maintenance actions under different distributions of initial deterioration conditions of roads and pipes.

4. Conclusion

This paper established a network-wide maintenance decision-making framework for joint long-term maintenance planning under uncertainties associated with deteriorating and co-located TI and WI systems with complex interdependencies. The framework captured uncertainties in road degradation and pipe failures under maintenance actions, and integrated traffic and hydraulic simulations to account for intra- and inter-system interdependencies between co-located roads and pipes. Based on the varied topological relationships among different co-located roads and pipes, their corresponding direct maintenance costs, as well as short-term and long-term user costs (reflecting traffic delays, vehicle depreciation, water loss, and service disruptions) resulting from maintenance activities, were comprehensively investigated. Due to the complexity of jointly maintaining network-wide interdependent TI and WI components, DRL was employed to identify optimal policies that minimize expected cumulative costs over a long-term planning horizon under future uncertainties. A realistic testbed of co-located TI and WI in Ybor City, Tampa, was developed to demonstrate the framework's cost-saving potential. This testbed compared the proposed strategy with conventional reactive and proactive maintenance approaches that overlooked TI-WI interdependencies. Results showed that the proposed strategy outperformed both baselines by explicitly incorporating TI-WI interdependencies into joint planning. By doing so, the framework

strategically adopted more proactive maintenance actions, achieving cost savings through joint interventions on co-located components, reducing redundant work, and avoiding major interventions such as road repaving. The approach also lowered cumulative costs over time by mitigating long-term user impacts associated with delayed maintenance. A comprehensive analysis was conducted to examine various factors, such as traffic volume, asset length, number of co-located components, and pipe failure risk levels, influence maintenance decisions. Furthermore, sensitivity analysis demonstrated that the strategy adapted to spatial deterioration patterns, prioritizing high-need areas to minimize long-term costs and disruptions.

There are several limitations of this work, and we plan to address them in future research. First, the current degradation modeling for roads and pipes mainly assumed stationary transition dynamics under relatively stable environmental and usage conditions. We also assumed that all deteriorating states were fully observable, such as road surface roughness index or pipe age. In future work, we will incorporate non-stationary transition dynamics to account for climate change impacts and develop more complex state representations, including partially observable condition states, such as deteriorating road structural health and hidden internal corrosion in pipes, where non-destructive testing or structural inspections are required and measurement errors must be considered. Second, the current reward functions did not capture the variability and fluctuation of cost parameters, nor did they explicitly

model detailed cost redistribution among different agencies. As future work, we will address uncertainties in cost parameter estimation due to limited data by incorporating a robust optimization framework. We will also investigate more detailed economic modeling of cost redistribution among different agencies, including cost-sharing agreements, financial responsibilities, and potential benefits for each infrastructure stakeholder. Third, our current case study focused on a relatively small testbed in Ybor City. In the future, we will scale up by collecting more data at the county or state level to examine the generalizability of the proposed maintenance model. We will also compare and evaluate opportunities to apply more advanced reinforcement learning methods and investigate their computational efficiencies.

CRedit authorship contribution statement

Bingjie Wang: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Formal analysis, Data curation, Conceptualization. **Hung Nguyen:** Writing – review & editing, Resources, Methodology, Conceptualization. **Qing Lu:** Writing – review & editing, Validation, Resources, Methodology, Data curation, Conceptualization. **Qiong Zhang:** Writing – review & editing, Validation, Resources, Methodology, Data curation, Conceptualization. **Mingyang Li:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Project administration, Methodology, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

- [1] US department of transportation - federal highway administration. Length by present serviceability rating (pavement condition)/measured pavement roughness—Major collectors and urban minor arterials and collectors (report no. HM-63). 2022.
- [2] Fan X, Wang X, Zhang X, Yu XB. Machine learning based water pipe failure prediction: The effects of engineering, geology, climate and socio-economic factors. *Reliab Eng Syst Saf* 2022;219:108185. <http://dx.doi.org/10.1016/j.res.2021.108185>.
- [3] Mazumder RK, Salman AM, Li Y, Yu X. Asset management decision support model for water distribution systems: Impact of water pipe failure on road and water networks. *J Water Resour Plan Manag* 2021;147(5):04021022, [https://ascelibrary.org/doi/full/10.1061/\(ASCE\)WR.1943-5452.0001365](https://ascelibrary.org/doi/full/10.1061/(ASCE)WR.1943-5452.0001365).
- [4] Aşçilean I, Iliescu M, Ciont N, Giurca I. The unfavourable impact of street traffic on water distribution pipelines. *Water* 2018;10(8):1086. <http://dx.doi.org/10.3390/w10081086>.
- [5] Hu Q, Ayinde O, Liu W. Mechanical response of buried water pipes to traffic loading before and after extreme cold waves. *Transp Geotech* 2024;49:101418. <http://dx.doi.org/10.1016/j.trgeo.2024.101418>.
- [6] Pericault Y, Viklander M, Hedström A. Modelling the long-term sustainability impacts of coordination policies for urban infrastructure rehabilitation. *Water Res* 2023;236:119912. <http://dx.doi.org/10.1016/j.watres.2023.119912>.
- [7] Xin J, Akiyama M, Frangopol DM. Sustainability-informed management optimization of asphalt pavement considering risk evaluated by multiple performance indicators using deep neural networks. *Reliab Eng Syst Saf* 2023;238:109448.
- [8] Rifaai TM, Abokifa AA, Sela L. Integrated approach for pipe failure prediction and condition scoring in water infrastructure systems. *Reliab Eng Syst Saf* 2022;220:108271.
- [9] Arismendi R, Barros A, Grall A. Piecewise deterministic Markov process for condition-based maintenance models—Application to critical infrastructures with discrete-state deterioration. *Reliab Eng Syst Saf* 2021;212:107540.
- [10] Wang W, Chen X. Piecewise deterministic Markov process for condition-based imperfect maintenance models. *Reliab Eng Syst Saf* 2023;236:109271.
- [11] Dekker R, Wildeman RE, der Duyn Schouten F.A. Van. A review of multi-component maintenance models with economic dependence. *Math Methods Oper Res* 1997;45(3):411–35.
- [12] Ramos-Salgado C, Muñuzuri J, Aparicio-Ruiz P, Onieva L. A decision support system to design water supply and sewer pipes replacement intervention programs. *Reliab Eng Syst Saf* 2021;216:107967.
- [13] Dui H, Chen S, Zhou Y, Wu S. Maintenance analysis of transportation networks by the traffic transfer principle considering node idle capacity. *Reliab Eng Syst Saf* 2022;221:108386.
- [14] Dinh DH, Do P, Jung B, Nguyen PTN. Reliability modeling and opportunistic maintenance optimization for a multicomponent system with structural dependence. *Reliab Eng Syst Saf* 2024;241:109708.
- [15] Yousefi N, Coit DW, Song S, Feng Q. Optimization of on-condition thresholds for a system of degrading components with competing dependent failure processes. *Reliab Eng Syst Saf* 2019;192:106547.
- [16] Li M, Wu B. Optimal condition-based opportunistic maintenance policy for two-component systems considering common cause failure. *Reliab Eng Syst Saf* 2024;250:110269.
- [17] Kheyri A, Taghipour S. Constrained optimal maintenance strategies for k-out-of-n systems with dependent components and mission duration. *Reliab Eng Syst Saf* 2025;111265.
- [18] Cheng W, Zhao X. Maintenance optimization for dependent two-component degrading systems subject to imperfect repair. *Reliab Eng Syst Saf* 2023;240:109581.
- [19] Zhang W, Zhang X, He S, Zhao X, He Z. Optimal condition-based maintenance policy for multi-component repairable systems with economic dependence in a finite-horizon. *Reliab Eng Syst Saf* 2024;241:109612.
- [20] Rinaldi SM, Peerenboom JP, T.K. Kelly. Identifying, understanding, and analyzing critical infrastructure interdependencies. *IEEE Control Syst Mag* 2001;21(6):11–25.
- [21] Kammouh O, Fecarotti C, Marandi A. A scalable optimization approach to the intervention planning of complex interconnected infrastructures. *Reliab Eng Syst Saf* 2024;250:110281.
- [22] Hu Q, Zhang Y. A Bi-Level Optimization Framework for Water Supply Network Repairs Considering Traffic Impact. *Water* 2025;17(6).
- [23] Nguyen H, Abdel-Mottaleb N, Uddin S, Zhang Q, Lu Q, Zhang H, Li M. Joint maintenance planning of deteriorating co-located road and water infrastructures with interdependencies. *Reliab Eng Syst Saf* 2022;226:108678.
- [24] Ilalokhoin O, Pant R, Hall JW. A model and methodology for resilience assessment of interdependent rail networks—Case study of Great Britain's rail network. *Reliab Eng Syst Saf* 2023;229:108895.
- [25] Kong J, Zhang C, Simonovic SP. Optimizing the resilience of interdependent infrastructures to regional natural hazards with combined improvement measures. *Reliab Eng Syst Saf* 2021;210:107538.
- [26] Xu M, Li G, Chen A. Resilience-driven post-disaster restoration of interdependent infrastructure systems under different decision-making environments. *Reliab Eng Syst Saf* 2024;241:109599.
- [27] Pei S, Zhai C, Hu J. Surrogate model-assisted seismic resilience assessment of the interdependent transportation and healthcare system considering a two-stage recovery strategy. *Reliab Eng Syst Saf* 2024;244:109941.
- [28] Sterling RL. Indirect costs of utility placement and repair beneath streets. In: Final report (report no. MN/rC-94-20). Minneapolis: Minnesota Department of Transportation; 1994.
- [29] Wichman C, Cunningham B. A cost of aging water infrastructure: traffic congestion resources. 2016, <https://www.resources.org/common-resources/a-cost-of-aging-water-infrastructure-traffic-congestion/>.
- [30] Abdel-Mottaleb N, Zhang Q. Water distribution—transportation interface connectivity responding to urban geospatial morphology. *J Infrastruct Syst* 2020;26(3):04020025.
- [31] Zhang N, Si W. Deep reinforcement learning for condition-based maintenance planning of multi-component systems under dependent competing risks. *Reliab Eng Syst Saf* 2020;203:107094.
- [32] Marugán AP, Pinar-Pérez JM, Márquez FPG. A reinforcement learning agent for maintenance of deteriorating systems with increasingly imperfect repairs. *Reliab Eng Syst Saf* 2024;252:110466.
- [33] Mohammadi R, He Q. A deep reinforcement learning approach for rail renewal and maintenance planning. *Reliab Eng Syst Saf* 2022;225:108615.
- [34] Hamida Z, Goulet JA. Hierarchical reinforcement learning for transportation infrastructure maintenance planning. *Reliab Eng Syst Saf* 2023;235:109214.
- [35] Saifullah M, Andriotis CP, Papakonstantinou KG, Stoffels SM. Deep reinforcement learning-based life-cycle management of deteriorating transportation systems. In: Bridge safety, maintenance, management, life-cycle, resilience and sustainability. CRC Press; 2022, p. 293–301.
- [36] Andriotis CP, Papakonstantinou KG. Deep reinforcement learning driven inspection and maintenance planning under incomplete information and constraints. *Reliab Eng Syst Saf* 2021;212:107551.
- [37] Morato PG, Andriotis CP, Papakonstantinou KG, Rigo P. Inference and dynamic decision-making for deteriorating systems with probabilistic dependencies through Bayesian networks and deep reinforcement learning. *Reliab Eng Syst Saf* 2023;235:109144.

- [38] Van Remmerden J, Kenter M, Roijers DM, Andriotis C, Zhang Y, Bukhsh Z. Deep multi-objective reinforcement learning for utility-based infrastructural maintenance optimization. *Neural Comput Appl* 2025;1–24.
- [39] Múčka P. International Roughness Index specifications around the world. *Road Mater Pavement Des* 2017;18(4):929–65.
- [40] Kleiner Y, Rajani B. Comprehensive review of structural deterioration of water mains: statistical models. *Urban Water* 2001;3(3):131–50.
- [41] Giraldo-González MM, Rodríguez JP. Comparison of statistical and machine learning models for pipe failure modeling in water distribution networks. *Water* 2020;12(4):1153.
- [42] Moerman A, Wols BA, Diemel R. The effects of traffic loads on drinking water main failure frequencies in the Netherlands. *Water Pr Technol* 2016;11(3):524–30.
- [43] Bukhsh ZA, Molegraaf H, Jansen N. A maintenance planning framework using online and offline deep reinforcement learning. *Neural Comput Appl* 2023;1–12.
- [44] Alvarez Lopez P, Behrisch M, Bieker-Walz L, Erdmann J, Flötteröd Y-P, Hilbrich R, Lücken L, Rummel J, Wagner P, Wießner E. Microscopic traffic simulation using SUMO. In: 2018 21st international conference on intelligent transportation systems. 2018, p. 2575–82. <http://dx.doi.org/10.1109/ITSC.2018.8569938>, Retrieved from <https://elib.dlr.de/124092/>.
- [45] Abdel-Mottaleb N, Saghand PG, Wakhungu MJ, Charkhgard H, Wells EC, Zhang Q. Identifying critical isolation valves in a water distribution network: A socio-technical approach. *Water* 2022;14(21):3587.
- [46] Abdel-Mottaleb N, Zhang Q. Logical architecture of water distribution networks. In: *World environmental and water resources congress 2019: hydraulics, waterways, and water distribution systems analysis*. Reston, VA: American Society of Civil Engineers; 2019, p. 506–19.
- [47] City of Tampa. City of tampa pre-approved products list. 2023, https://www.tampa.gov/sites/default/files/document/2023/city_of_tampa_pre-approved_products_list_0.pdf. [Accessed 24 Aug 2025].
- [48] City of Tampa. Historic ybor. 2024, <https://www.tampa.gov/ybor-city>. [Accessed 24 Aug 2025].
- [49] OpenStreetMap contributors. Planet dump, OpenStreetMap 2023. <https://www.openstreetmap.org>. [Accessed 24 Aug 2025].
- [50] J. Stigloher, J. Erdmann, M. Behrisch. OsmWebWizard.py script [software]. Eclipse SUMO. 2014, <https://eclipse.dev/sumo>. [Accessed 24 Aug 2025].
- [51] Florida Department of Transportation (FDOT). Annual average daily traffic (AADT) records. 2023, <https://tdaappsprod.dot.state.fl.us/fto/>. [Accessed 2024].
- [52] Cruz OGD, Mendoza CA, Lopez KD. International roughness index as road performance indicator: a literature review. In: *IOP conference series: earth and environmental science*. vol. 822, (1):IOP Publishing; 2021, 012016.
- [53] Folkman S. Pvc pipe longevity report: Affordability and the 100+ year benchmark standard. 2014.
- [54] Clark RM, Sivaganesan M, Selvakumar A, Sethi V. Cost models for water supply distribution systems. *J Water Resour Plan Manag* 2002;128(5):312–21.
- [55] Mallela J, Sadavisam S. Work zone road user costs: Concepts and applications. 2011.
- [56] Lu Q, Li M, Gunaratne M, Xin C, Hoque M, Rajalingola M. Best practices for construction and repair of bridge approaches and departures. 2018.
- [57] Ahopelto S, Vahala R. Cost-benefit analysis of leakage reduction methods in water supply networks. *Water* 2020;12(1):195.
- [58] Zhang S, Li H, Wang M, Liu M, Chen PY, Lu S, et al. On the convergence and sample complexity analysis of deep Q-networks with ϵ -greedy exploration. *Adv Neural Inf Process Syst* 2023;36:13064–102.
- [59] De La Fuente N, Guerra DAV. A comparative study of deep reinforcement learning models: DQN vs PPO vs A2C. 2024, arXiv preprint [arXiv:2407.14151](https://arxiv.org/abs/2407.14151).
- [60] C. McCray. Ybor City Redevelopment Area. 2025, <https://www.tampa.gov/CRA/ybor-city>. [Accessed 24 Aug 2025].